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V9938 MSX-VIDEO

Technical Data Book

ASCII CORPORATION / NIPPON GAKKI CO., LTD.

Transcribed by Laurens Holst

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PREFACE

The V9938 introduced in this manual is a Very Large-Scale Integrated Circuit (VLSI) that was developed as a Video Display Processor (VDP) for the MSX2. The MSX personal computer standard was introduced in 1983 by ASCII Corporation and Microsoft Incorporated. At present, the MSX is manufactured and marketed worldwide. In 1985, out of the desire to strengthen some of the functions of the original MSX, the MSX2 standard was developed. In addition to being software-compatible with the MSX, the MSX2 supports new media and has video processing capabilities that are not available on conventional 8-bit personal computers.

To make the MSX2 a reality, two requirements for the Video Processor were upward compatibility with the existing TMS9918A (the VDP for the MSX) software while increasing the number of functions. The V9938 was developed through the joint efforts of ASCII Corporation, Microsoft Incorporated, and YAMAHA.

The following functions are supported on the V9938.

- Full bit-mapped mode
- 80-column text display
- Access using X- and Y-coordinates. The load of the I/O driver has been lightened. The X-Y coordinates are independent of the screen mode.
- Fundamental commands implemented by hardware to decrease the processing time of the I/O driver: AREA MOVE, LINE, SEARCH, RASTER OPERATION, etc.
- Digitize and external synchronization
- Color palette (9 bits x 16 patterns)
- Linear RGB video output
- More sprites per horizontal line

Because the V9938 has the above functions, it provides for superior video capabilities that make it possible for its use in a variety of applications, including the MSX2. CAPTAIN terminals and NAPLPS terminals using the V9938 have already been developed. We hope that the V9938 will be a standard video processing device on a worldwide basis.

This manual was written so as to explain how to set the parameters of the V9938 and is a reference for developing applications and systems software for it.

We are pleased that you have chosen to develop software for the V9938 and that you have referred to this manual for assistance.

Finally, we would like to express our deep gratitude to the people at NTT as well as the other related manufacturers for their valuable opinions which contributed to the development of the v9938.

August, 1985

ASCII Corporation

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PART 1

MSX-VIDEO DATA PROCESSOR V9938 USER'S MANUAL

BASIC INPUT AND OUTPUT

1. Accessing the Control Registers

There are two ways to set data in the MSX-VIDEO control registers (R#0 to R#46), which we will describe below.

1.1 Direct access

Output the data and the register number in sequence to port #1. Since this order is always used, be careful when you access the MSX-VIDEO for an interrupt routine.

	MSB	7	6	5	4	3	2	1	0	LSB	
Port #1 First byte		D7	D6	D5	D4	D3	D2	D1	D0		DATA
Second byte		1	0	R5	R4	R3	R2	R1	R0		REGISTER #

1.2 Indirect access

Specify the register number in control register R#17 (Control Register Pointer).

First set the register number in R#17 (using direct addressing) by sending data to port #3. When you set the data in R#17, you can also set its MSB (AII, the autoincrement bit) to control autoincrementing. The data in R#17 cannot be changed by indirect addressing.

If autoincrement is prohibited, the contents of R#17 will be unchanged, and thus you do not have to reset R#17.

	MSB	7	6	5	4	3	2	1	0	LSB	
Register #17		AII	0	R5	R4	R3	R2	R1	R0		REGISTER #
		└─ 1: Auto increment inhibit └─ 0: Auto increment on									
Port #3 First byte		D7	D6	D5	D4	D3	D2	D1	D0		DATA
Port #3 Second byte		D7	D6	D5	D4	D3	D2	D1	D0		DATA
		...									
Port #3 nth byte		D7	D6	D5	D4	D3	D2	D1	D0		DATA

2. Accessing the Palette Registers

To set data in the MSX-VIDEO palette registers (P#0 to P#15/9 bit), you must first set the palette register number in register R#16 (Color palette address pointer) and subsequently output the two bytes of data (in order) through port #2.

	MSB	7	6	5	4	3	2	1	0	LSB	
Register #16		0	0	0	0	C3	C2	C1	C0		Palette #
Port #2 First byte		0	R2	R1	R0	0	B2	B1	B0		DATA
		Red data					Blue data				
Port #2 Second byte		0	0	0	0	0	G2	G1	G0		DATA
							Green data				

(Editor's note: R#16 increases after every second data byte written)

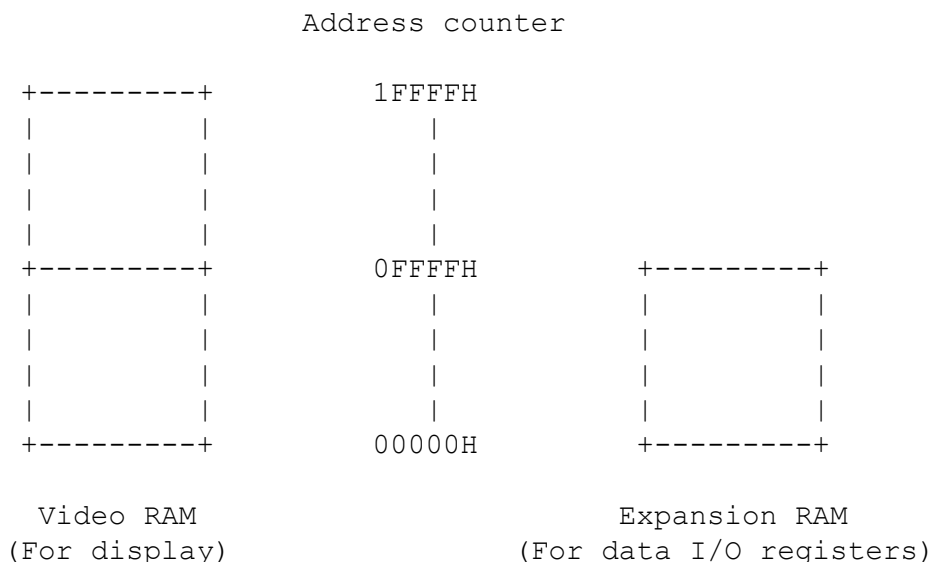
3. Accessing the Status Registers

To read the status registers of the MSX-VIDEO (S#0 to S#9), you must first set the register number in R#15 (Status register pointer) and read the data through Port #1.

	MSB	7	6	5	4	3	2	1	0	LSB	
Register #15		0	0	0	0	S3	S2	S1	S0		Status register
Port #1 Read data		D7	D6	D5	D4	D3	D2	D1	D0		DATA

4. Accessing the Video RAM

A Video RAM of 128K bytes plus an expansion RAM of 64K bytes can be connected to the MSX-VIDEO. The memory maps for these cases are shown in the map below.



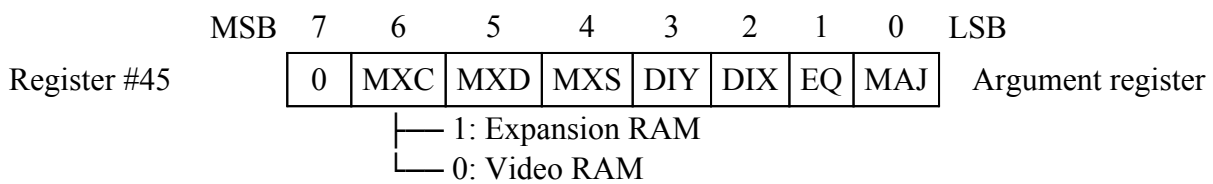
Accessing memory

To access memory, follow the procedure below.

1. Switch banks (VRAM to Expansion RAM)
2. Set the address counter (A16 to A14)
3. Set the address counter (A7 to A0)
4. Set the address counter (A13 to A8), and specify read or write
5. Read or write the data

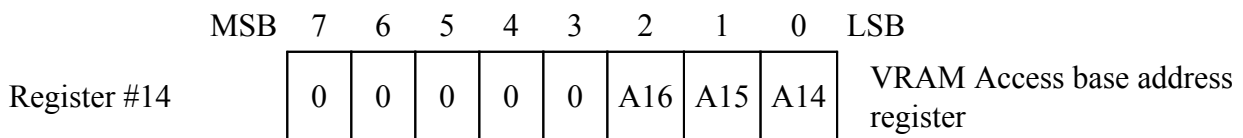
1. Switching banks (VRAM to Expansion RAM)

Since the contents of R#45 (Argument register) do not change each time that memory is accessed, it is not necessary to respecify bit 6 of register R#45 (which specifies banking) every time that you are to do banking.



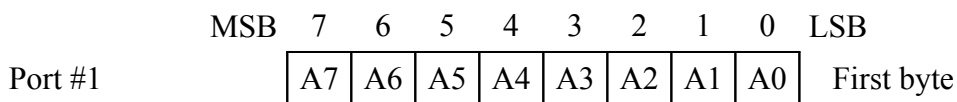
2. Setting the address counter (A16-A14)

Set the high-order three bits (A16 to A14) of the address counter using register R#14 (VRAM Access base address register).



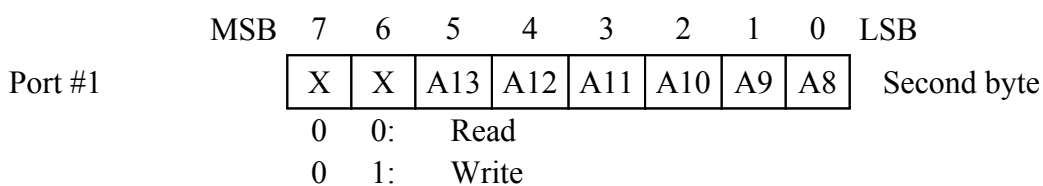
3. Setting the address counter (A7 to A0)

Set the low-order eight bits (A7 to A0) of the address counter by outputting data to Port #1.



4. Setting the address counter (A13 to A8) and specifying read or write

Set the remaining six bits (A13 to A8) of the address counter and specify read or write by outputting data to Port #1.



5. Reading or writing data

Since the address counter is automatically incremented when data is read from or written to Port #0, you may continually access blocks of data.

- To access the VRAM, you can also use commands. These commands will be explained in a later chapter.
- Refer to the data sheet for access timings.

REGISTER FUNCTIONS

1. CONTROL REGISTERS

#0 to #23 (Write only)

#32 to #46 (Write only)

1.1 Mode Registers

	MSB	7	6	5	4	3	2	1	0	LSB
R#0		0	DG	IE2	IE1	M5	M4	M3	0	Mode Register 0
R#1		0	BL	IE0	M1	M2	0	SI	MAG	Mode Register 1
R#8		MS	LP	TP	CB	VR	0	SPD	BW	Mode Register 2
R#9		LN	0	S1	S0	IL	EO	$\overline{\text{NT}}$	DC	Mode Register 3

R#0 DG: Sets the color bus to input mode, and inputs data into the VRAM.

IE2: Enables interrupt from Lightpen by Interrupt Enable 2.

IE1: Enables interrupt from Horizontal scanning line by Interrupt Enable 1.

M5: Used to change the display mode.

M4: Used to change the display mode.

M3: Used to change the display mode.

R#1 BL: When 1, screen display enabled. When 0, screen disabled.

IE0: Enables interrupt from Horizontal scanning line by Interrupt Enable 0.

M1: Used to change the display mode.

M2: Used to change the display mode.

SI: When 1, sprite size is 16 x 16. When 0, 8 x 8.

MA: Sprite expansion; when 1: expanded. When 0, normal.

R#8 MS: When 1, sets the color bus to input mode and enables mouse.
When 0, sets the color bus to output mode and disables mouse.

LP: When 1, enables light pen. When 0, disables light pen.

TP: Sets the color of code 0 to the color of the palette.

CB: When 1, sets the color bus to input mode.

When 0, sets the color bus to output mode.

VR: Selects the type of Video RAM.

1 = 64K x 1 bit or 64K x 4 bits.

0 = 16K x 1 bit or 16K x 4 bits.

SPD: When 1, disables display of sprites. When 0, displays sprites.

BW: When 1, sets black and white in 32 tones.

When 0, sets color (available only with a composite encoder).

R#9 LN: When 1, sets the horizontal dot count to 212.
When 0, sets the horizontal dot count to 192.

S1: Selects simultaneous mode.

S0: Selects simultaneous mode.

IL: When 1, interlace (Complete NTSC timing)
When 0, non-interlace (Incomplete NTSC timing)

EO: When 1, displays two graphic screens interchangeably by Even field/Odd field.
When 0, displays the same graphic screen by Even field/Odd field.

$\overline{\text{NT}}$: When 1, PAL (313 lines); when 0, NTSC (262 lines). (For RGB output only)

DC: When 1, sets *DLCLK to input mode; when 0, sets *DLCLK to output mode.

1.2 Table Base Address Registers

The table base address registers are a set of registers to declare the addresses of tables in the VRAM to be used by MSX-VIDEO.

Note that when these registers are accessed, the control codes that the screen may receive depends on the display mode. For this purpose, you must mask the unwanted bits.

	MSB	7	6	5	4	3	2	1	0	LSB	
R#2		0	A16	A15	A14	A13	A12	A11	A10		Pattern name table base address register
R#3		A13	A12	A11	A10	A9	A8	A7	A6		Color table base address register low
R#10		0	0	0	0	0	A16	A15	A14		Color table base address register high
R#4		0	0	A16	A15	A14	A13	A12	A11		Pattern generator table base address register
R#5		A14	A13	A12	A11	A10	A9	A8	A7		Sprite attribute table base address register low
R#11		0	0	0	0	0	0	A16	A15		Sprite attribute table base address register high
R#6		0	0	A16	A15	A14	A13	A12	A11		Sprite pattern generator table base address register

1.3 Color Registers

The color registers are used to control the MSX-VIDEO's text and background screen colors as well as blinking, etc.

```

      MSB  7    6    5    4    3    2    1    0  LSB
      +---+---+---+---+---+---+---+---+
R#7   |TC3|TC2|TC1|TC0|BD3|BD2|BD1|BD0| Text color/Back drop
      +---+---+---+---+---+---+---+---+ color register

      TC3 to TC0:  Specifies the text color according to TEXT 1
                    and TEXT 2 modes.
      BD3 to BD0:  Specifies the back drop color in all display
                    modes.
      MSB  7    6    5    4    3    2    1    0  LSB
      +---+---+---+---+---+---+---+---+
R#12  |T23|T22|T21|T20|BC3|BC2|BC1|BC0| Text color/Back color
      +---+---+---+---+---+---+---+---+ register

```

In TEXT 2 mode, if the attributes for blinking are set, the color set in this register and set in R#7 are displayed alternately.

```

      T23 to T20:  Specifies the color of part 1 of the pattern.
      BC3 to BC0:  Specifies the color of part 0 of the pattern.

      MSB  7    6    5    4    3    2    1    0  LSB
      +---+---+---+---+---+---+---+---+
R#13  |ON3|ON2|ON1|ON0|OF3|OF2|OF1|OF0| Blinking period register
      +---+---+---+---+---+---+---+---+

```

In the bit map modes of GRAPH4 to GRAPH7, the two pages are alternately displayed (blinked). Place data in this register to set the display page to an odd page to begin blinking. This register is also used in the TEXT2 mode.

```

      ON3 to ON0:  Display time for even page
      OF3 to OF0:  Display time for odd page

      MSB  7    6    5    4    3    2    1    0  LSB
      +---+---+---+---+---+---+---+---+
R#20  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Color burst register 1
      +---+---+---+---+---+---+---+---+
R#21  | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | Color burst register 2
      +---+---+---+---+---+---+---+---+
R#22  | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | Color burst register 3
      +---+---+---+---+---+---+---+---+

```

The above values are preset when the power is applied. If all values in the above three registers are set to 0, the color burst signal of the composite video output will be erased.

If the above values are subsequently reset to the preset values, the normal color burst signal will be output.

1.4 Display Registers

The display registers are used to control the display position on the CRT.

```

      MSB  7    6    5    4    3    2    1    0  LSB
      +---+---+---+---+---+---+---+---+
R#18  |V3 |V2 |V1 |V0 |H3 |H2 |H1 |H0 | Display adjust register
      +---+---+---+---+---+---+---+---+

```

The above register is used to adjust the display position on the CRT.

```

H = 7 . . . H = 1,  H = 0,  H = 15 . . . H = 8
(Left)                (Center)                (Right)

V = 8 . . . V = 15, V = 0,  V = 1 . . . V = 7
(Bottom)              (Center)              (Top)

```

(Editor's note: modifying this value while executing a VDP command will result in a corrupted byte)

```

      MSB  7    6    5    4    3    2    1    0  LSB
      +---+---+---+---+---+---+---+---+
R#23  |DO7|DO6|DO5|DO4|DO3|DO2|DO1|DO0| Display offset register
      +---+---+---+---+---+---+---+---+

```

The above register sets the location of the line to begin display.

```

      VRAM (Line)

TOP  +-----+ 0
    |         |
    | CRT Screen |
    |         |
BOTTOM +-----+ 192 or 212  R#23 = 0
    |         |
    +-----+ 255

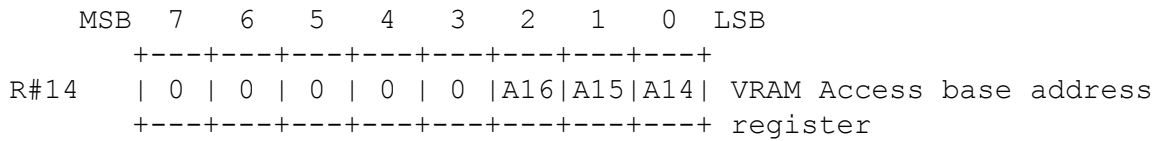
    +-----+ 0
    |         |
    | CRT Screen |
    |         |
BOTTOM +-----+ 136 or 156  R#23 = 200
    |         |
TOP    +-----+ 200
    | CRT Screen |
    +-----+ 255
MSB  7    6    5    4    3    2    1    0  LSB
+---+---+---+---+---+---+---+---+
R#19 |IL7|IL6|IL5|IL4|IL3|IL2|IL1|IL0| Interrupt line register
+---+---+---+---+---+---+---+---+

```

You may specify interrupts when the MSX-VIDEO begins to display a specified scanning line. To enable the interrupt, use the above register to set the scanning line.

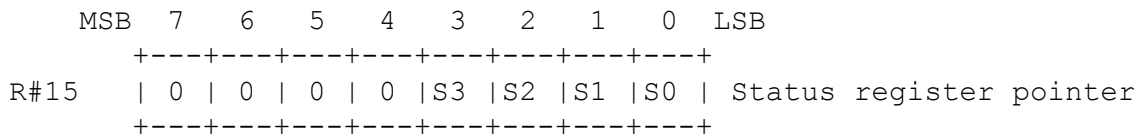
1.5 ACCESS REGISTERS

The access registers are a set of registers used when accessing the MSX-VIDEO registers or the VRAM.

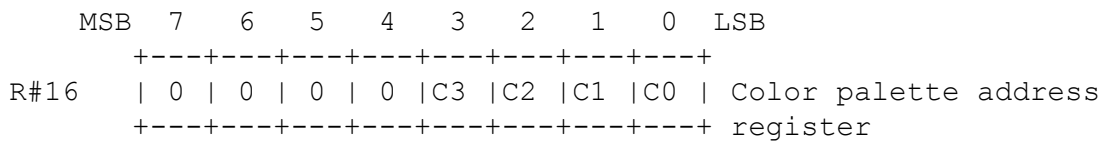


When accessing the MSX-VIDEO and the Video RAM (VRAM), set the high-order three bits of the address in the VRAM access base address register.

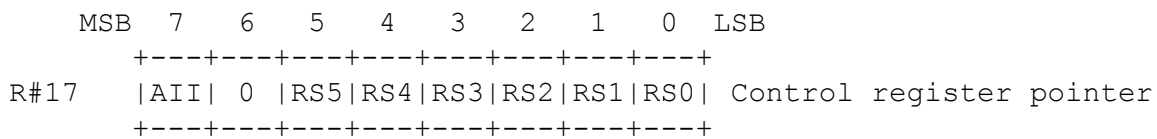
When data is set in this register, and the VRAM is accessed, if there is a carry from A13, the data in the register is automatically incremented. In GRAPHIC1, GRAPHIC2, MULTICOLOR, and TEXT1 modes, the data in the register is not automatically incremented.



When reading the MSX-VIDEO status registers (S#0 to S#9), set the contents of the status register pointer.



When setting the color palette of the MSX-VIDEO, set the number of the palette in the Color palette address register.



In the MSX-VIDEO, the above control register pointer may be used to access another register. In addition, according to the setting of the AII bit, the data can be automatically incremented.

AII = 1: Auto increment disabled
AII = 0: Auto increment enabled

1.6 COMMAND REGISTERS

The following command registers are used when executing a command on the MSX-VIDEO. Details on the use of these command registers will be presented in a later chapter.

	MSB	7	6	5	4	3	2	1	0	LSB								
		+	---	+	---	+	---	+	---	+								
R#32		SX7		SX6		SX5		SX4		SX3		SX2		SX1		SX0		Source X low register
		+	---	+	---	+	---	+	---	+								
R#33		0		0		0		0		0		0		0		SX8		Source X high register
		+	---	+	---	+	---	+	---	+								
R#34		SY7		SY6		SY5		SY4		SY3		SY2		SY1		SY0		Source Y low register
		+	---	+	---	+	---	+	---	+								
R#35		0		0		0		0		0		0		SY9		SY8		Source Y high register
		+	---	+	---	+	---	+	---	+								
		+	---	+	---	+	---	+	---	+								
R#36		DX7		DX6		DX5		DX4		DX3		DX2		DX1		DX0		Destination X low register
		+	---	+	---	+	---	+	---	+								
R#37		0		0		0		0		0		0		0		DX8		Destination X high register
		+	---	+	---	+	---	+	---	+								
R#38		DY7		DY6		DY5		DY4		DY3		DY2		DY1		DY0		Destination Y low register
		+	---	+	---	+	---	+	---	+								
R#39		0		0		0		0		0		0		DY9		DY8		Destination Y high register
		+	---	+	---	+	---	+	---	+								
		+	---	+	---	+	---	+	---	+								
R#40		NX7		NX6		NX5		NX4		NX3		NX2		NX1		NX0		Number of dots X low register
		+	---	+	---	+	---	+	---	+								
R#41		0		0		0		0		0		0		0		NX8		Number of dots X high register
		+	---	+	---	+	---	+	---	+								
R#42		NY7		NY6		NY5		NY4		NY3		NY2		NY1		NY0		Number of dots Y low register
		+	---	+	---	+	---	+	---	+								
R#43		0		0		0		0		0		0		NY9		NY8		Number of dots Y high register
		+	---	+	---	+	---	+	---	+								
		+	---	+	---	+	---	+	---	+								
R#44		CH3		CH2		CH1		CH0		CL3		CL2		CL1		CL0		Color register
		+	---	+	---	+	---	+	---	+								
R#45		0		MXC		MXD		MXS		DIY		DIX		EQ		MAJ		Argument register
		+	---	+	---	+	---	+	---	+								
R#46		CM3		CM2		CM1		CM0		LO3		LO2		LO1		LO0		Command register
		+	---	+	---	+	---	+	---	+								

2. STATUS REGISTERS #0 to #9 (Read only)

The following status registers are read-only registers for reporting the status when the MSX-VIDEO is read.

```

      MSB  7    6    5    4    3    2    1    0    LSB
      +---+---+---+---+---+---+---+---+
S#0    | F |5S | C |Fifth sprite number| Status register 0
      +---+---+---+---+---+---+---+---+
F:     Vertical scanning interrupt flag
      When S#0 is read, this flag is reset.

5S:    Flag for the fifth sprite
      Five sprites are aligned on the first horizontal line (In
      the G3 to G7 modes, 9 sprites are allowed)

C:     Collision flag
      Two sprites have collided.

Fifth sprite number:
      The number of the fifth (or ninth) sprite.

      MSB  7    6    5    4    3    2    1    0    LSB
      +---+---+---+---+---+---+---+---+
S#1    |FL |LPS| Identification #   |FH | Status register 1
      +---+---+---+---+---+---+---+---+
FL:    Lightpen flag (Lightpen flag set)
      If the lightpen is to detect light, this bit as well as the
      IE2 bit must both set in order for an interrupt to be
      enabled. When S#1 is read, FL is reset.

      Mouse switch 2 (Mouse flag set)
      The second switch on the mouse was pressed.
      In this case, when S#1 is read, FL is not reset.

LPS:   Lightpen switch (Lightpen flag set)
      The lightpen switch was pressed.
      In this case, when S#1 is read, LPS is not reset.

      Mouse switch 1 (Mouse flag set)
      The first switch on the mouse was pressed.
      In this case, when S#1 is read, LPS is not reset.

Identification number:
      The identification number (ID #) of the MSX-VIDEO.
      (Editor's note: 0 if V9938, 1 if V9948, 2 if V9958)

FH:    Horizontal scanning interrupt flag
      Horizontal scanning interrupt (which is specified in R#19)
      flag. If IE1 is set, an interrupt is enabled. When S#1
      is read, FH is reset.

      MSB  7    6    5    4    3    2    1    0    LSB
      +---+---+---+---+---+---+---+---+
S#2    |TR |VR |HR |BD | 1 | 1 |EO |CE | Status register 2
      +---+---+---+---+---+---+---+---+
TR:    Transfer ready flag
      When the CPU sends commands to the VRAM and other devices,
      the CPU checks this flag while transferring data. When this
      flag is set to 1, transfer may be done.

```

VR: Vertical scanning line timing flag
During vertical scanning, this flag is set to 1.

HR: Horizontal scanning line timing flag
During horizontal scanning, this flag is set to 1.

BD: Boundary color detect flag
When the search command is executed, this flag detects whether the boundary color was detected or not.

EO: Display field flag
When 0, indicates the first field.
When 1, indicates the second field.

CE: Command execution flag
Indicates that a command is being executed.

	MSB	7	6	5	4	3	2	1	0	LSB	
		+	---	+	---	+	---	+	---	+	
S#3		X7	X6	X5	X4	X3	X2	X1	X0	Column register low	
		+	==	+	==	+	==	+	==	+	
S#4		1	1	1	1	1	1	1	X8	Column register high	
		+	==	+	==	+	==	+	==	+	
S#5		Y7	Y6	Y5	Y4	Y3	Y2	Y1	Y0	Row register low	
		+	==	+	==	+	==	+	==	+	
S#6		1	1	1	1	1	1	EO	Y8	Row register high	
		+	---	+	---	+	---	+	---	+	

The above registers are set to indicate the collision location of sprites, the location of lightpen detection, and the relative movement of the mouse.

	MSB	7	6	5	4	3	2	1	0	LSB	
		+	---	+	---	+	---	+	---	+	
S#7		C7	C6	C5	C4	C3	C2	C1	C0	Color register	
		+	---	+	---	+	---	+	---	+	

The above color register is used when the POINT and VRAM to CPU commands are executed. The VRAM data is set in this register.

	MSB	7	6	5	4	3	2	1	0	LSB	
		+	---	+	---	+	---	+	---	+	
S#8		BX7	BX6	BX5	BX4	BX3	BX2	BX1	BX0	Border X register low	
		+	==	+	==	+	==	+	==	+	
S#9		1	1	1	1	1	1	1	BX8	Border X register high	
		+	---	+	---	+	---	+	---	+	

When the search command is executed and the border color has been detected, the X coordinate is set in the above registers.

TEXT 1 MODE

Characteristics

- Pattern size : 6 dots (w) x 8 dots (h)
- Patterns : 256 types
- Screen pattern count : 40 (w) x 24 (h) patterns
- Pattern colors : Two colors out of 512 colors (per screen)
- VRAM area per screen : 4K bytes

Controls

- Pattern font : VRAM pattern generator table
- Screen pattern location : VRAM pattern name table
- Pattern color code 1 : High-order four bits of R#7
- Pattern color code 0 : Low-order four bits of R#7
- Background color code : Low-order four bits of R#7

Initial settings

1. Mode and register settings

	MSB	7	6	5	4	3	2	1	0	LSB	
R#0		0	DG	IE2	IE1	0*	0*	0*	0		Mode register 0
R#1		0	BL	IE0	1*	0*	0	SI	MAG		Mode register 1
R#8		MS	LP	TP	CB	VR	0	SPD	BW		Mode register 2
R#9		LN	0	S1	S0	IL	EO	**NT	DC		Mode register 3

* Examples of settings in TEXT 1 mode

** Indicates negative logic

All other bits are set accordingly

2. Pattern Generator Table Settings

- The pattern generator table is an area that stores the pattern fonts.
- Each pattern had a number from PN0 to PN255.
- The font for each pattern is constructed from 8 bytes, and the lower two bits of each of the eight bytes is not displayed.
- Set the beginning (head) address of the pattern generator table is register R#4.

```

      MSB  7    6    5    4    3    2    1    0    LSB
      +---+---+---+---+---+---+---+---+
R#4    | 0 | 0 | A16|A15|A14|A13|A12|A11| Pattern generator table
      +---+---+---+---+---+---+---+---+ base address register

```

Pattern generator table

(X=1, 0=0)

```

      ++----- These bits are not displayed.
      ||
MSB 76543210 LSB
0 00X00000 -----+
1 0X0X0000         |
2 X000X000         |
3 X000X000         +---- Pattern number 0
4 XXXXX000         |
5 X000X000         |
6 X000X000         |
7 00000000 -----+
8 XXXX0000 -----+
9 X000X000         |
10 X000X000        |
11 XXXX0000        +---- Pattern number 1
12 X000X000        |
13 X000X000        |
14 XXXX0000        |
15 00000000 -----+
.      .      .
.      .      .
.      .      .
2040 X0X0X000 -----+
2041 0X0X0X00      |
2042 X0X0X000      |
2043 0X0X0X00      +---- Pattern number 0
2044 X0X0X000      |
2045 0X0X0X00      |
2046 X0X0X000      |
2047 0X0X0X00 -----+

```

3. Pattern name table settings

- The pattern name table is composed of one byte for each screen pattern. Each byte specifies a unique pattern.
- Set the beginning (head) address of the pattern name table in register R#2.

```

      MSB  7    6    5    4    3    2    1    0    LSB
      +---+---+---+---+---+---+---+---+
R#2    | 0 | A16|A15|A14|A13|A12|A11|A10| Pattern name table
      +---+---+---+---+---+---+---+---+ base address register

```

Pattern name table

[illegible]

4. Color register settings

	MSB	7	6	5	4	3	2	1	0	LSB	
	+---+---+---+---+---+---+---+---+										
R#7	TC3	TC2	TC1	TC0	BD3	BD2	BD1	BD0	Text color/Back drop color		
	+---+---+---+---+---+---+---+---+										register
					+---+---+---+---+					Specifies pattern color	
										code 0 or back drop color	
	+---+---+---+---+---+---+---+---+										Specifies pattern color
											code 1

Example of VRAM allocation in TEXT 1 mode

00000H	+-----+	MSB	7	6	5	4	3	2	1	0	LSB
	Pattern										
	Name	R#2	0	0	0	0	0	0	0	0	
	Table 0										
				A16	A15	A14	A13	A12	A11	A10	
003C0H	+-----+										
00800H	+-----+	MSB	7	6	5	4	3	2	1	0	LSB
	Pattern										
	Generator	R#4	0	0	0	0	0	0	0	1	
	Table 0										
				A16	A15	A14	A13	A12	A11		
01000H	+-----+	MSB	7	6	5	4	3	2	1	0	LSB
	Pattern										
	Name	R#2	0	0	0	0	0	1	0	0	
	Table 1										
				A16	A15	A14	A13	A12	A11	A10	
013C0H	+-----+										
01800H	+-----+	MSB	7	6	5	4	3	2	1	0	LSB
	Pattern										
	Generator	R#4	0	0	0	0	0	0	1	1	
	Table 1										
				A16	A15	A14	A13	A12	A11		
02000H	+-----+										
	. .										
	. .										
	. .										
	. .										
1FFFFH	+-----+										

A maximum of 32 pages may be allocated in the same manner (using a 128K-byte VRAM).

TEXT 2 MODE

Characteristics

- Pattern size : 6 dots (w) x 8 dots (h)
- Patterns : 256 types
- Screen pattern count : 80 (w) x 24 (h) patterns
80 (w) x 26.5 (h) patterns
- Pattern blinking : Possible for each character
- Pattern colors : Two colors out of 512 colors (per screen),
four if using blinking
- VRAM area per screen : 8K bytes

Controls

- Pattern font : VRAM pattern generator table
- Screen pattern location : VRAM pattern name table
- Blink attributes : VRAM color table
- Pattern color code 1 : High-order four bits of R#7
- Pattern color code 0 : Low-order four bits of R#7
- Background color code : Low-order four bits of R#7
- Pattern color code 1 : High-order four bits of R#12
(used for blinking)
- Pattern color code 0 : Low-order four bits of R#12
(used for blinking)

Initial Settings

1. Mode and Register Settings

	MSB	7	6	5	4	3	2	1	0	LSB								
	+---+---+---+---+---+---+---+---+---+---+																	
R#0		0		DG		IE2		IE1		0*		1*		0*		0		Mode register 0
	+===+===+===+===+===+===+===+===+===+===+																	
R#1		0		BL		IE0		1*		0*		0		SI		MAG		Mode register 1
	+===+===+===+===+===+===+===+===+===+===+																	
R#8		MS		LP		TP		CB		VR		0		SPD		BW		Mode register 2
	+===+===+===+===+===+===+===+===+===+===+																	
R#9		LN		0		S1		S0		IL		EO		**NT		DC		Mode register 3
	+---+---+---+---+---+---+---+---+---+---+																	

* Examples of settings in TEXT 2 mode

** Indicates negative logic

In this display mode, if LN is set to 1, 26.5 lines are selected, and if LN is set to 0, 24 lines are selected.

All other bits are set accordingly.

2. Pattern Generator Table Settings

- The pattern generator table is an area that stores the pattern fonts.
- Each pattern has a number from PN0 to PN255.
- Set the beginning (head) address of the pattern generator table in register R#4

```

      MSB  7    6    5    4    3    2    1    0    LSB
      +---+---+---+---+---+---+---+---+
R#4   | 0 | 0 | A16|A15|A14|A13|A12|A11| Pattern generator table
      +---+---+---+---+---+---+---+---+ base address register

```

- The font for each pattern is constructed from 8 bytes, and the lower two bits of each of the eight bytes is not displayed.

Pattern generator table

(X=1, 0=0)

```

      ++----- These bits are not displayed.
      ||
MSB 76543210 LSB
  0 OOX00000 -----+
  1 OXOX0000         |
  2 X000X000         |
  3 X000X000         +---- Pattern number 0
  4 XXXXX000         |
  5 X000X000         |
  6 X000X000         |
  7 00000000 -----+
  8 XXXX0000 -----+
  9 X000X000         |
10 X000X000         |
11 XXXX0000         +---- Pattern number 1
12 X000X000         |
13 X000X000         |
14 XXXX0000         |
15 00000000 -----+
.      .      .
.      .      .
.      .      .
2040 XOXOX000 -----+
2041 OXOXOX00         |
2042 XOXOX000         |
2043 OXOXOX00         +---- Pattern number 255
2044 XOXOX000         |
2045 OXOXOX00         |
2046 XOXOX000         |
2047 OXOXOX00 -----+

```

3. Pattern name table settings

- The pattern name table is composed of one byte for each screen pattern. Each byte specifies a unique pattern.
- If LN is set to 0, the screen display pattern is 80 (W) x 24 (H); and if LN is set to 1, the screen display pattern is 80 (W) x 26.5 (H). The upper half of the 27th pattern is displayed.
- Set the beginning (head) address of the pattern name table in register R#2.

```

      MSB  7    6    5    4    3    2    1    0    LSB
      +---+---+---+---+---+---+---+---+
R#2   | 0 |A16|A15|A14|A13|A12| 1 | 1 | Pattern name table
      +---+---+---+---+---+---+---+---+ base address register

```

Pattern name table

```

|          |
+-----+ Base address
| ( 0, 0) | 0          0    1    2    3    .    .    79    X
+-----+
| ( 1, 0) | 1          0 | 0 | 1 | 2 | 3 | .    . | 79 |
+-----+
| ( 2, 0) | 2          1 | 80 | 81 | 82 | 83 | .    . | 159 |
+-----+
.          .    .          .    .    .    .    .    .
.          .    .          .    .    .    .    .    .
.          .    .          +---+---+          --+---+
+-----+          25 |2000|2001| .    .    .    . |2079|
| (79, 0) | 79          +---+---+          --+---+
+-----+          26 |2080|2081| .    .    .    . |2159|
| ( 0, 1) | 80          +---+---+          --+---+
+-----+          Y
.          .    .
.          .    .          Screen display correspondence
.          .    .
+-----+
| (79,26) | 2159
+-----+

```

4. Color table settings

- In TEXT 2 mode, each pattern has a separate bit for the attribute area, and if this bit is set to 1, the pattern blink attribute will be set.
- Set the beginning (head) address of the color table in registers R#3 and R#10.

```

      MSB  7    6    5    4    3    2    1    0    LSB
      +---+---+---+---+---+---+---+---+
R#3   |A13|A12|A11|A10|A9 | 1 | 1 | 1 | Color table
      +---+---+---+---+---+---+---+---+
R#10  | 0 | 0 | 0 | 0 | 0 |A16|A15|A14| base address registers
      +---+---+---+---+---+---+---+---+

```

Color table

MSB	7	6	5	4	3	2	1	0	LSB
+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
0	(0, 0}	(1, 0}	(2, 0}	(3, 0}	(4, 0}	(5, 0}	(6, 0}	(7, 0}	Base address
+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
1	(8, 0}	(9, 0}	(10, 0}	(11, 0}	(12, 0}	(13, 0}	(14, 0}	(15, 0}	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
.	.	.	.	.	.	.	.	.	
.	.	.	.	.	.	.	.	.	
.	.	.	.	.	.	.	.	.	
.	.	.	.	.	.	.	.	.	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
269	(72, 26}	.	.	.	.	.	.	(79, 26}	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+									

5. Color register settings

- Set the color for pattern 1 in the high-order bits of register R#7.
- Set the color for pattern 0 in the low-order bits of register R#7.

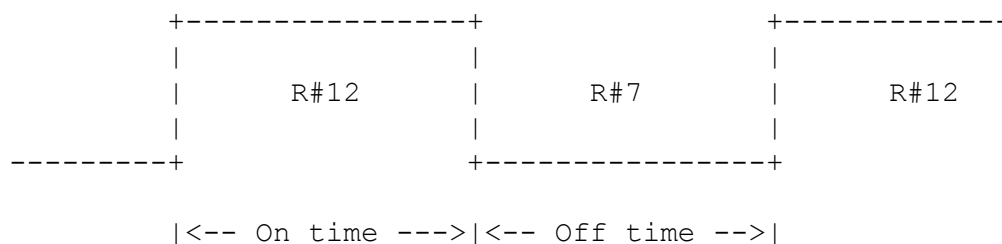
MSB	7	6	5	4	3	2	1	0	LSB
+---+---+---+---+---+---+---+---+---+									
R#7	TC3	TC2	TC1	TC0	BD3	BD2	BD1	BD0	Text color/Back drop color register
+---+---+---+---+---+---+---+---+---+									

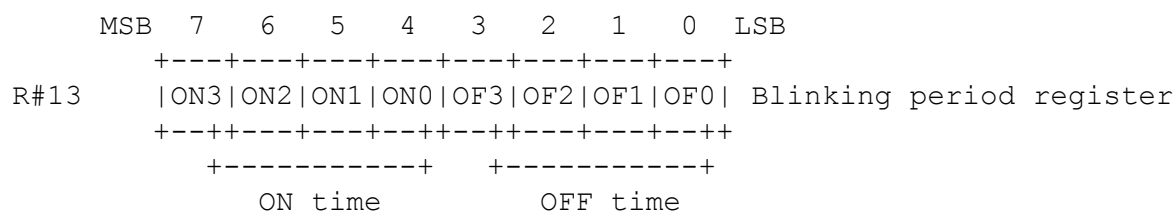
- Set the blink attribute for the corresponding pattern by setting an alternate color code in register R#12. The pattern will be blinked by using the color codes set in register R#7 and R#12.

MSB	7	6	5	4	3	2	1	0	LSB
+---+---+---+---+---+---+---+---+---+									
R#12	T23	T22	T21	T20	BC3	BC2	BC1	BC0	Text color/Back color register
+---+---+---+---+---+---+---+---+---+									

6. Blink register settings

- The color codes set in registers R#7 and R#12 will be alternately displayed for blinking; however, the blinking period attribute (time on and time off) can also be set in register R#13.





- The data for the ON and OFF times are shown below. (NTSC)

DATA (binary)	TIME (ms)	VBlanks
0 0 0 0	0	0
0 0 0 1	166.9	10
0 0 1 0	333.8	20
0 0 1 1	500.5	30
0 1 0 0	667.5	40
0 1 0 1	843.4	50
0 1 1 0	1001.3	60
0 1 1 1	1168.2	70
1 0 0 0	1335.1	80
1 0 0 1	1501.9	90
1 0 1 0	1668.8	100
1 0 1 1	1835.7	110
1 1 0 0	2002.6	120
1 1 0 1	2169.5	130
1 1 1 0	2336.3	140
1 1 1 1	2503.2	150

Example of VRAM allocation in TEXT 2 mode

		Pattern name table base address									
00000H	+-----+	MSB	7	6	5	4	3	2	1	0	LSB
	Pattern	+---+---+---+---+---+---+---+---+---+									
	Name	R#2	0	0	0	0	0	0	1	1	
	Table 0	+---+---+---+---+---+---+---+---+---+									
00870H	+-----+	A16 A15 A14 A13 A12									
00A00H	+ Color -+	Color table base address									
	Table 0										
00B0EH	+-----+										
00800H	+-----+	MSB	7	6	5	4	3	2	1	0	LSB
	Pattern	+---+---+---+---+---+---+---+---+---+									
	Generator	R#3	0	0	1	0	1	1	1	1	
	Table 0	+---+---+---+---+---+---+---+---+---+									
01800H	+-----+	A13 A12 A11 A10 A9									
	Pattern	+---+---+---+---+---+---+---+---+---+									
	Generator	R#10	0	0	0	0	0	0	0	0	
	Table 0	+---+---+---+---+---+---+---+---+---+									
02000H	+-----+	A16 A15 A14									
	Pattern										
	Name										
	Table 1	Pattern generator table base address									
02870H	+-----+										
02A00H	+ Color -+	MSB	7	6	5	4	3	2	1	0	LSB
	Table 1	+---+---+---+---+---+---+---+---+---+									
02B0EH	+-----+	R#4	0	0	0	0	0	0	1	0	
03000H	+-----+	+---+---+---+---+---+---+---+---+---+									
	Pattern	A16 A15 A14 A13 A12 A11									
	Generator										
	Table 0										
03800H	+-----+										
04000H	+-----+										
	.										
	.										
	.										
	.										
1FFFFH	+-----+										

A maximum of 16 pages may be allocated in the same manner (using a 128K-byte VRAM).

MULTICOLOR MODE

Characteristics

- Screen composition : 64 (w) x 48 (h) color blocks
- Color blocks : Sixteen colors out of 512 colors
- Sprite mode : Sprite mode 1
- VRAM area per screen : 4K bytes

Controls

- Color block color code : VRAM pattern generator table
- Color block location : VRAM pattern name table
- Background color code : Low-order four bits of R#7
- Sprites : VRAM sprite attribute table
VRAM sprite pattern table

Initial Settings

1. Mode and Register Settings

	MSB	7	6	5	4	3	2	1	0	LSB	
R#0		0	DG	IE2	IE1	0*	0*	0*	0		Mode register 0
R#1		0	BL	IE0	0*	1*	0	SI	MAG		Mode register 1
R#8		MS	LP	TP	CB	VR	0	SPD	BW		Mode register 2
R#9		LN	0	S1	S0	IL	EO	**NT	DC		Mode register 3

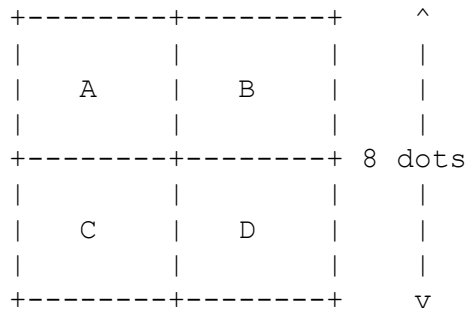
* Examples of settings in MULTICOLOR mode

** Indicates negative logic

2. Pattern Generator Table Settings

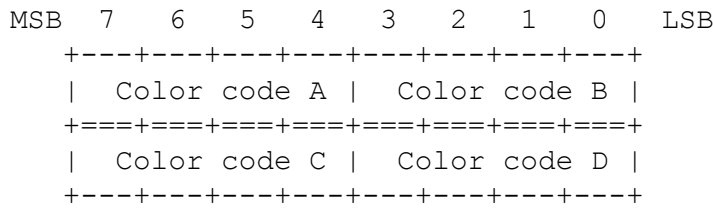
- The pattern generator table is an area that stores the colors of the color blocks.
- Each pattern is made up of four color blocks. These patterns are approximately 8 x 8 when the dots available for the screen display area is 256 x 192 dots.

<----- 8 dots ---->

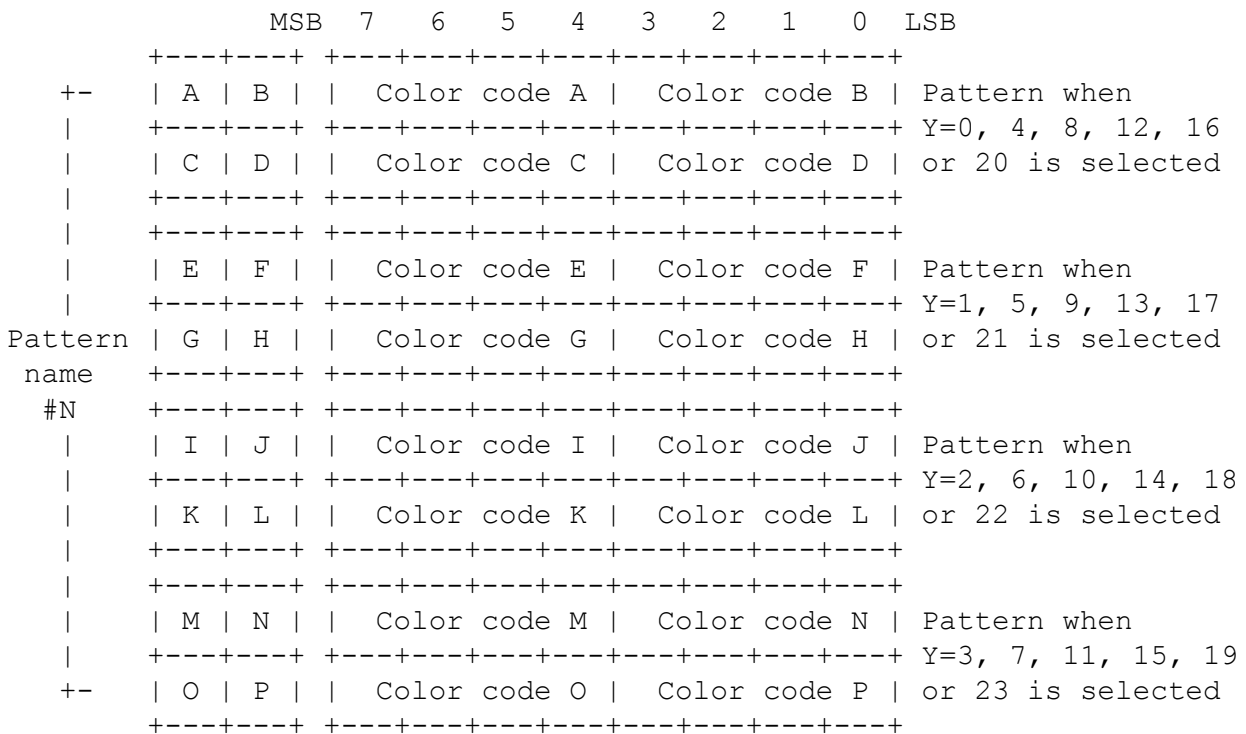


For each block A, B, C, and D, sixteen colors may be specified.

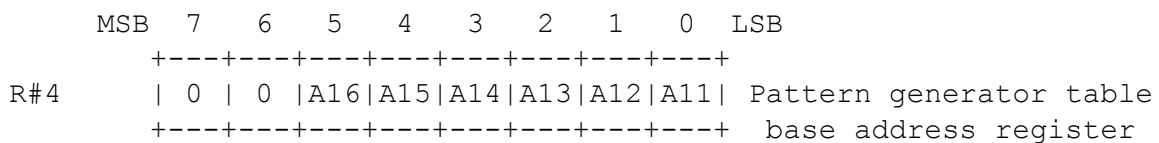
- In the MULTICOLOR mode, two bytes are used for each pattern, and each pattern includes four color blocks.



- In the MULTICOLOR mode, for each pattern name, there are four corresponding color blocks, and according to the y-coordinate, the pattern names are automatically set.



- Set the beginning (head) address of the pattern generator table in register R#4.



Pattern generator table

+-----+		0 Base address
Pattern name #0		
(Eight bytes)		
+-----+	8	
Pattern name #1		
(Eight bytes)		
+-----+	16	
.	.	
.	.	
.	.	
+-----+	2040	
Pattern name #255		
(Eight bytes)		
+-----+	2048	

3. Pattern name table settings

- The pattern name table is composed of one byte for each screen pattern. Each byte specifies a unique pattern number.

Pattern name table

	0	1	2	3	...	31
0	0	1	2	3	...	31
1	32	33	34	35	...	63
...	...	...	...	...	...	...
...	...	...	...	...	...	...
22	704	705	...	...	...	735
23	736	737	...	...	...	767

- Set the beginning (head) address of the pattern name table in register R#2.

	MSB	7	6	5	4	3	2	1	0	LSB	
R#2		0	A16	A15	A14	A13	A12	A11	A10		Pattern name table base address register

Pattern name table

+-----+	Base address	
(0, 0)	0	
+-----+		
(1, 0)	1	
+-----+		
(2, 0)	2	
+-----+		
.	.	.
.	.	.
.	.	.
+-----+		
(31, 0)	31	
+-----+		
(0, 1)	32	
+-----+		
.	.	.
.	.	.
.	.	.
+-----+		
(31,23)	767	
+-----+		

4. Color register settings

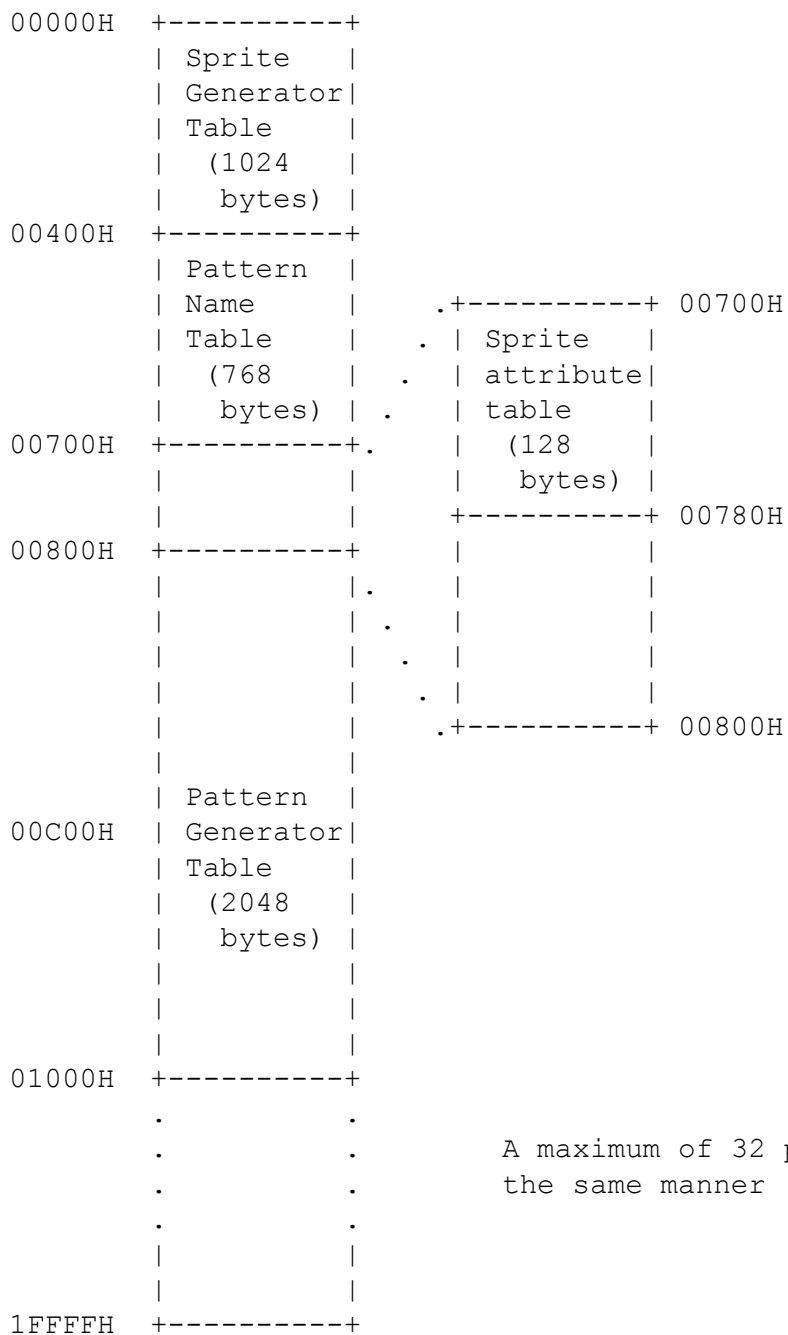
	MSB	7	6	5	4	3	2	1	0	LSB	
		+---	+---	+---	+---	+---	+---	+---	+---		
R#7		TC3	TC2	TC1	TC0	BD3	BD2	BD1	BD0		Text color/Back drop color
		+--+	+--+	+--+	+--+	+--+	+--+	+--+	+--+		register
						+---	+---	+---	+---		Specifies backdrop color
		+---	+---	+---	+---	-----	-----	-----	-----		Ignored

5. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 1.

	MSB	7	6	5	4	3	2	1	0	LSB	
		+---	+---	+---	+---	+---	+---	+---	+---		
R#5		A14	A13	A12	A11	A10	A9	A8	A7		Sprite attribute table
		+===+	+===+	+===+	+===+	+===+	+===+	+===+	+===+		base address register
R#11		0	0	0	0	0	0	A16	A15		
		+---	+---	+---	+---	+---	+---	+---	+---		
		+---	+---	+---	+---	+---	+---	+---	+---		
R#6		0	0	A16	A15	A14	A13	A12	A11		Sprite pattern generator
		+---	+---	+---	+---	+---	+---	+---	+---		table base address
											register

Example of VRAM allocation in MULTICOLOR mode



A maximum of 32 pages may be allocated in the same manner (using a 128K-byte VRAM).

GRAPHIC 1 MODE

Characteristics

- Pattern size : 8 dots (w) x 8 dots (h)
- Patterns : 256 types
- Screen pattern count : 32 (w) x 24 (h) patterns
- Pattern colors : 16 colors out of 512 colors (per screen)
- Sprite mode : Sprite mode 1
- VRAM area per screen : 4K bytes

Controls

- Pattern font : VRAM pattern generator table
- Screen pattern location : VRAM pattern name table
- Pattern color codes 1 & 0: Can be specified as a group for each 8-pattern set, in the VRAM color table
- Background color code : Low-order four bits of R#7
- Sprites : VRAM sprite attribute table, VRAM sprite pattern table

Initial Settings

1. Mode and Register Settings

	MSB	7	6	5	4	3	2	1	0	LSB								
	+---+---+---+---+---+---+---+---+---+---+																	
R#0		0		DG		IE2		IE1		0*		0*		0*		0		Mode register 0
	+===+===+===+===+===+===+===+===+===+===+																	
R#1		0		BL		IE0		0*		0*		0		SI		MAG		Mode register 1
	+===+===+===+===+===+===+===+===+===+===+																	
R#8		MS		LP		TP		CB		VR		0		SPD		BW		Mode register 2
	+===+===+===+===+===+===+===+===+===+===+																	
R#9		LN		0		S1		S0		IL		EO		**NT		DC		Mode register 3
	+---+---+---+---+---+---+---+---+---+---+																	

* Examples of settings in GRAPHIC 1 mode

** Indicates negative logic

2. Pattern Generator Table Settings

- The pattern generator table is an area that stores the pattern fonts.
- Each pattern has a number from PN0 to PN255.
- The font for each pattern is constructed from 8 bytes.
- Set the beginning (head) address of the pattern generator table in register R#4.

	MSB	7	6	5	4	3	2	1	0	LSB	
		+---+---+---+---+---+---+---+---+									
R#4		0	0	A16	A15	A14	A13	A12	A11		Pattern generator table
		+---+---+---+---+---+---+---+---+								base address register	

Pattern generator table

(X=1, O=0)

	MSB	7	6	5	4	3	2	1	0	LSB	
		+---+---+---+---+---+---+---+---+									
0		O	O	X	X	X	O	O	O		-----+
1		O	X	O	O	O	X	O	O		
2		X	O	O	O	O	X	O			
3		X	O	O	O	O	X	O			+--- Pattern number 0
4		X	X	X	X	X	X	X	O		
5		X	O	O	O	O	X	O			
6		X	O	O	O	O	X	O			
7		O	O	O	O	O	O	O	O		-----+
8		X	X	X	X	X	X	X	O		-----+
9		X	O	O	O	O	X	O			
10		X	O	O	O	O	X	O			
11		X	X	X	X	X	X	X	O		+--- Pattern number 1
12		X	O	O	O	O	X	O			
13		X	O	O	O	O	X	O			
14		X	X	X	X	X	X	X	O		
15		O	O	O	O	O	O	O	O		-----+
.		.	.	.	.	.	.	.	.		.
.		.	.	.	.	.	.	.	.		.
.		.	.	.	.	.	.	.	.		.
2040		X	O	X	O	X	O	X	O		-----+
2041		O	X	O	X	O	X	O	X		
2042		X	O	X	O	X	O	X	O		
2043		O	X	O	X	O	X	O	X		+--- Pattern number 255
2044		X	O	X	O	X	O	X	O		
2045		O	X	O	X	O	X	O	X		
2046		X	O	X	O	X	O	X	O		
2047		O	X	O	X	O	X	O	X		-----+

3. Pattern name table settings

- The pattern name table is composed of one byte for each screen pattern. Each byte specifies a unique pattern.
- Set the beginning (head) address of the pattern name table in register R#2.

	MSB	7	6	5	4	3	2	1	0	LSB	
		+---+---+---+---+---+---+---+---+									
R#2		0	A16	A15	A14	A13	A12	A11	A10		Pattern name table
		+---+---+---+---+---+---+---+---+								base address register	

Pattern name table

[illegible]

4. Color register settings

	MSB	7	6	5	4	3	2	1	0	LSB	
		+---+---+---+---+---+---+---+---+									
R#7		TC3	TC2	TC1	TC0	BD3	BD2	BD1	BD0		Text color/Back drop color
		+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+								register	
						+---+---+---+---				Specifies backdrop color	
		+---+---+---+-----								Ignored	

5. Color table settings

- The colors for pattern color 1 and pattern color 0 are set in groups of eight patterns.
- Set the beginning (head) address of the color table in registers R#3 and R#10.

```

      MSB  7    6    5    4    3    2    1    0  LSB
R#3      +---+---+---+---+---+---+---+---+
          |A13|A12|A11|A10|A9 |A8 |A7 |A6 | Color table
          +===+===+===+===+===+===+===+===+
R#10     | 0 | 0 | 0 | 0 | 0 |A16|A15|A14| base address register
          +---+---+---+---+---+---+---+---+

```

Color table

[illegible]

6. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 1.

	MSB	7	6	5	4	3	2	1	0	LSB	
	+---+---+---+---+---+---+---+---+										
R#5	A14 A13 A12 A11 A10 A9 A8 A7										Sprite attribute table
	+===+===+===+===+===+===+===+===+										base address register
R#11	0 0 0 0 0 0 A16 A15										
	+---+---+---+---+---+---+---+---+										
	+---+---+---+---+---+---+---+---+										
R#6	0 0 A16 A15 A14 A13 A12 A11										Sprite pattern generator
	+---+---+---+---+---+---+---+---+										table base address
											register

Example of VRAM allocation in GRAPHIC 1 mode

```

00000H +-----+
        | Sprite   |
        | Generator|
        | Table     |
        | (1024    |
        | bytes)   |
00400H +-----+
        | Pattern  |
        | Name      | .+-----+ 00700H
        | Table     | . | Sprite   |
        | (768      | . | attribute|
        | bytes)    | . | table    |
00700H +-----+. | (128      |
        |           | | bytes)   |
        |           | +-----+ 00780H
00800H +-----+ | Color     |
        |           | | table    |
        |           | . + (32 bytes)+ 007A0H
        |           | . |         |
        |           | . |         |
        |           | .+-----+ 00800H
        |           |
        | Pattern  |
00C00H | Generator|
        | Table     |
        | (2048    |
        | bytes)   |
        |           |
        |           |
01000H +-----+
        .           .
        .           .
        .           .
        .           .
        |           |
        |           |
1FFFFH +-----+

```

A maximum of 32 pages may be allocated in the same manner (using a 128K-byte VRAM).

GRAPHIC 2 AND GRAPHIC 3 MODES

Characteristics

- Pattern size : 8 dots (w) x 8 dots (h)
- Patterns : 768 types
- Screen pattern count : 32 (w) x 24 (h) patterns
- Pattern colors : 16 colors out of 512 colors (per screen)
- Sprite modes : Sprite mode 1 (GRAPHIC 2)
: Sprite mode 2 (GRAPHIC 3)
- Vram area per screen : 16K bytes

* The GRAPHIC 2 and GRAPHIC 3 modes are identical except for the sprite modes.

Controls

- Pattern font : VRAM pattern generator table
- Screen pattern location : VRAM pattern name table
- Pattern color codes 1 & 0: Can be specified as a group for each
: raster, in the VRAM color table
- Background color code : Low-order four bits of R#7
- Sprites : VRAM sprite attribute table, VRAM sprite pattern table.

Initial Settings

1. Mode and Register Settings

	MSB	7	6	5	4	3	2	1	0	LSB		
		+	---	+	---	+	---	+	---	+	---	+
R#0		0	DG	IE2	IE1	0*	0*	0*	0		Mode register 0	
		+	---	+	---	+	---	+	---	+	---	+
									+	-----	0, 1 for GRAPHIC 2 mode	
								+	-----		1, 0 for GRAPHIC 3 mode	
		+	---	+	---	+	---	+	---	+	---	+
R#1		0	BL	IE0	0*	0*	0	SI	MAG		Mode register 1	
		+	---	+	---	+	---	+	---	+	---	+
R#8		MS	LP	TP	CB	VR	0	SPD	BW		Mode register 2	
		+	---	+	---	+	---	+	---	+	---	+
R#9		LN	0	S1	S0	IL	EO	**NT	DC		Mode register 3	
		+	---	+	---	+	---	+	---	+	---	+

* Examples of settings in GRAPHIC 2 mode or GRAPHIC 3 mode

** Indicates negative logic

All other bits are set accordingly.

2. Pattern Generator Table Settings

- The pattern generator table is an area that stores the pattern fonts.
- Each pattern has a number from PN0 to PN255; and since each group may have three members, 768 patterns may be specified.
- The font for each pattern is constructed from 8 bytes.
- Set the beginning (head) address of the pattern generator table in register R#4.

	MSB	7	6	5	4	3	2	1	0	LSB								
	+---+---+---+---+---+---+---+---+																	
R#4		0		0		A16		A15		A14		A13		1		1		Pattern generator table
	+---+---+---+---+---+---+---+---+										base address register							

3. Color table settings

- The colors for pattern color 1 and pattern color 0 are set as a group of one raster.
- The color table corresponds to the pattern generator table on a one- to-one basis.
- Set the beginning (head) address of the color table in registers R#3 and R#10.

	MSB	7	6	5	4	3	2	1	0	LSB								
	+---+---+---+---+---+---+---+---+																	
R#3		A13		1		1		1		1		1		1		1		Color table address
	+---+---+---+---+---+---+---+---+																	
R#10		0		0		0		0		0		A16		A15		A14		register
	+---+---+---+---+---+---+---+---+																	

4. Color register settings

	MSB	7	6	5	4	3	2	1	0	LSB								
	+---+---+---+---+---+---+---+---+																	
R#7		TC3		TC2		TC1		TC0		BD3		BD2		BD1		BD0		Text color/Back drop color
	+---+---+---+---+---+---+---+---+										register							
							+---+---+---+---				Specifies backdrop color							
	+---+---+---+---+---+---+---+---+										Ignored							

Pattern generator table

(X=1, O=0)

		MSB 76543210 LSB	MSB 76543210 LSB
Pattern number 0	-----+	0 OOXXX000	00000000
		1 OX000X00	00000000
		2 X00000X0	00000000
	-----+	3 X00000X0	00000000
		4 XXXXXXXX0	00000000
		5 X00000X0	00000000
		6 X00000X0	00000000
	-----+	7 00000000	00000000
Pattern number 0	-----+	8 XXXXXXXX0	00000000
		9 X00000X0	00000000
		10 X00000X0	00000000
	-----+	11 XXXXXXXX0	00000000
		12 X00000X0	00000000
		13 X00000X0	00000000
		14 XXXXXXXX0	00000000
	-----+	15 00000000	00000000
	.	.	.
	.	.	.
	.	.	.
Pattern number 255	-----+	2040 XOXOXOXO	00000000
		2041 OXOXOXOX	00000000
		2042 XOXOXOXO	00000000
	-----+	2043 OXOXOXOX	00000000
		2044 XOXOXOXO	00000000
		2045 OXOXOXOX	00000000
		2046 XOXOXOXO	00000000
	-----+	2047 OXOXOXOX	00000000
	Color pattern 1	-----+	---+
	Color pattern 0	-----+	---+

Pattern generator
table

Color table

0					
	-----+	-----+	-----+	-----+	-----+
	256 patterns	256 colors			
	for upper	for upper			
	third of	third of			
	screen	screen			
	(2048 bytes)	(2048 bytes)			
2048	-----+	-----+	-----+	-----+	-----+
	256 patterns	256 colors			
	for middle	for middle			
	third of	third of			
	screen	screen			
	(2048 bytes)	(2048 bytes)			
4096	-----+	-----+	-----+	-----+	-----+
	256 patterns	256 colors			
	for lower	for lower			
	third of	third of			
	screen	screen			

	(2048 bytes)	(2048 bytes)	
6144	+-----+	+-----+	-+

5. Pattern name table settings

- The pattern name table is composed of one byte for each screen pattern. Each byte specifies a unique pattern.
- The upper, middle, and lower parts of the screen can be used as three different parts, for a total of 768 patterns.

Pattern name table

(0, 0)	Pattern display area for upper third of screen (256 bytes)	(31, 0)
(0, 7)		(31, 7)
(0, 8)	Pattern display area for middle third of screen (256 bytes)	(31, 8)
(0,15)		(31,15)
(0,16)	Pattern display area for lower third of screen (256 bytes)	(31,16)
(0,23)		(31,23)

- Set the beginning (head) address of the pattern name table in register R#2.

	MSB	7	6	5	4	3	2	1	0	LSB	
	+---	+---	+---	+---	+---	+---	+---	+---	+---		
R#2	0	A16	A15	A14	A13	A12	A11	A10		Pattern name table	
	+---	+---	+---	+---	+---	+---	+---	+---		base address register	

Pattern name table

		Base address	
(0, 0)	0		
(1, 0)	1		For upper third of screen
.	.	.	
(31, 7)	255		
(0, 8)	256		For middle third of screen
.	.	.	
(31,15)	511		
(0,16)	512		For lower third of screen
.	.	.	
(31,23)	767 (byte)		

6. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites pertaining to GRAPHIC 2 MODE, see the section on SPRITE MODE 1, and for details about sprites pertaining to GRAPHIC 3 mode, see the section on SPRITE MODE 2.

	MSB	7	6	5	4	3	2	1	0	LSB											
		+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---		
R#5			A14		A13		A12		A11		A10		A9		A8		A7			Sprite attribute table	
		+	===	+	===	+	===	+	===	+	===	+	===	+	===	+	===	+	===	+	base address register
R#11			0		0		0		0		0		0		A16		A15				
		+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	
		+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	
R#6			0		0		A16		A15		A14		A13		A12		A11			Sprite pattern generator	
		+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	table base address
		+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	register

Example of VRAM allocation in GRAPHIC 2 mode

00000H	+-----+				
		Pattern			
		Generator			
		Table			
		Upper			
00800H	+-----+				
		Pattern			
		Generator			
		Table			
		Middle			
01000H	+-----+				
		Pattern		.+-----+ 01C00H	
		Generator		. Sprite	
		Table		. generator	
		Lower		. table	
01800H	+-----+	.			
				+-----+ 01C00H	
02000H	+-----+			Sprite	
		Color		. attribute	
		Table		. table	
		Upper		. +-----+ 01C80H	
02800H	+-----+	.			
		Color		.	
		Table		.	
		Middle		+-----+ 02000H	
	+-----+				
03000H		Color			
		Table			
		Lower			
03800H	+-----+				
		Pattern			
		Name			
		Table			
03B00H	+-----+				
		Table			
04000H	+-----+				
	.	.			
	.	.			
	.	.			
	.	.			
1FFFFH	+-----+				

A maximum of 8 pages may be allocated in the same manner (using a 128K-byte VRAM).

Example of VRAM allocation in GRAPHIC 3 mode

00000H	+-----+				
	Pattern				
	Generator				
	Table				
	Upper				
00800H	+-----+				
	Pattern				
	Generator				
	Table				
	Middle				
01000H	+-----+				
	Pattern	.	+-----+	01C00H	
	Generator	.	Sprite		
	Table	.	generator		
	Lower	.	table		
01800H	+-----+	.			
			+-----+	01C00H	
02000H	+-----+		Sprite		
	Color .		color		
	Table .		table		
	Upper .		+-----+	01E00H	
02800H	+-----+	.	Sprite		
	Color .		attribute		
	Table .		table		
	Middle .		+-----+	01E80H	
	+-----+	.			
03000H	Color .				
	Table .		+-----+	02000H	
	Lower				
03800H	+-----+				
	Pattern				
	Name				
	Table				
03B00H	+-----+				
	Table				
04000H	+-----+				
	.	.			
	.	.			
	.	.			
	.	.			
1FFFFH	+-----+				

A maximum of 8 pages may be allocated in the same manner (using a 128K-byte VRAM).

GRAPHIC 4 MODE

Characteristics

- Bit-mapped Graphics Mode
- Screen size : 256 (w) x 212 (h) dots
: 256 (w) x 192 (h) dots
- Screen colors : 16 colors out of 512 colors (per screen)
- Sprite mode : Sprite mode 2
- VRAM area per screen : 32K bytes

Controls

- Graphics : VRAM pattern name table
- Background color code : Low-order four bits of R#7
- Sprites : VRAM sprite attribute table, VRAM sprite pattern table

Initial Settings

1. Mode and Register Settings

	MSB	7	6	5	4	3	2	1	0	LSB
	+---+---+---+---+---+---+---+---+---+---+									
R#0		0		DG		IE2 IE1	0*		1*	1* 0 Mode register 0
	+===+===+===+===+===+===+===+===+===+===+									
R#1		0		BL		IE0	0*		0*	0 SI MAG Mode register 1
	+===+===+===+===+===+===+===+===+===+===+									
R#8		MS		LP		TP		CB		VR 0 SPD BW Mode register 2
	+===+===+===+===+===+===+===+===+===+===+									
R#9		LN		0		S1		S0		IL EO **NT DC Mode register 3
	+---+---+---+---+---+---+---+---+---+---+									

* Examples of settings in GRAPHIC 4 mode

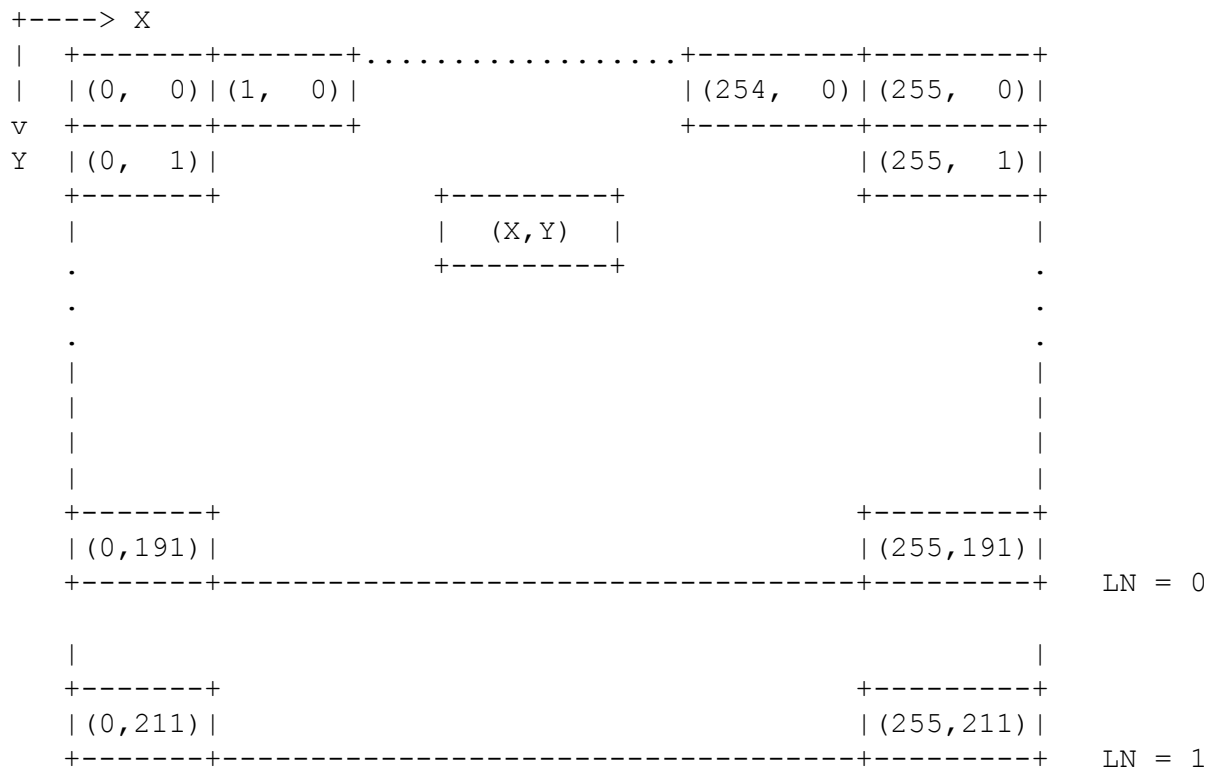
** Indicates negative logic

In GRAPHIC 4 mode, if LN is set to 1, the screen height is 212 dots, and if LN is set to 0, the screen height is 192 dots.

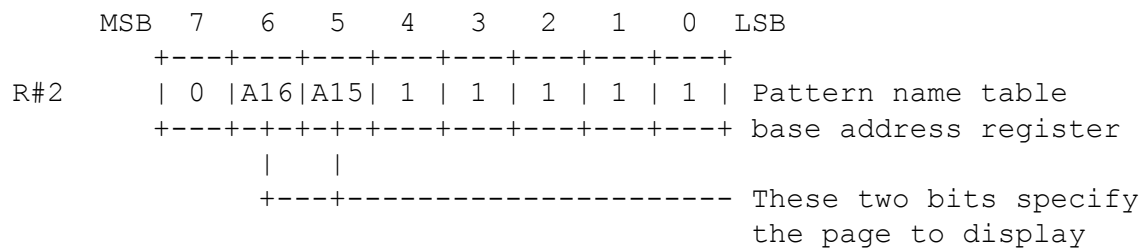
All other bits are set accordingly.

2. Pattern name table settings

- The pattern name table is composed of one byte for every two dots on the screen. A color can be assigned for each dot from a collection of 16 colors out of 512 colors.



- Set the beginning (head) address of the pattern name table in register R#2.



Pattern name table

MSB	7	6	5	4	3	2	1	0	LSB
	+---+---+---+---+---+---+---+---+								
1		(0, 0)				(1, 0)			Base address
	+---+---+---+---+---+---+---+---+								
2		(2, 0)				(3, 0)			Set the color code for each
	+---+---+---+---+---+---+---+---+								dot in this table
	.								.
	.								.
	.								.
	+---+---+---+---+---+---+---+---+								
127		(254, 0)				(255, 0)			
	+---+---+---+---+---+---+---+---+								
128		(0, 1)				(1, 1)			
	+---+---+---+---+---+---+---+---+								
	.								.
	.								.
	.								.
	+---+---+---+---+---+---+---+---+								
27134		(252, 211)				(253, 211)			
	+---+---+---+---+---+---+---+---+								
27135		(254, 211)				(255, 211)			
	+---+---+---+---+---+---+---+---+								

3. Color register settings

	MSB	7	6	5	4	3	2	1	0	LSB	
	+---+---+---+---+---+---+---+---+---+										
R#7		TC3	TC2	TC1	TC0	BD3	BD2	BD1	BD0		Text color/Back drop
	+---+---+---+---+---+---+---+---+---+										color register
					+---+---+---+---+---						Specifies backdrop color
											code
	+---+---+---+---+---+---+---+---+---									Ignored	

4. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 2.

	MSB	7	6	5	4	3	2	1	0	LSB	
		+	-	-	+	-	-	+	-	-	+
R#5		A14	A13	A12	A11	A10	1		1		1
		+	=	=	+	=	=	+	=	=	+
R#11			0		0		0		0		A16 A15
		+	-	-	+	-	-	+	-	-	+
		+	-	-	+	-	-	+	-	-	+
R#6			0		0		A16 A15	A14	A13	A12	A11
		+	-	-	+	-	-	+	-	-	+
		+	-	-	+	-	-	+	-	-	+

Example of VRAM allocation in GRAPHIC 4 mode

00000H	+	-----	+		
02000H		Pattern			
		name			
		table			
04000H					
		192 lines			
06000H	+	-----	+		
				.	+

		212 lines		.	+
				.-	Sprite
06A00H	+	-----	+		generator
				.	table
					(2048
07000H	+	-----	+		bytes)
					+

					07800H
					Sprite
					color
					table
					(512
					bytes)
					+

					07A00H
					Sprite
					attribute
					table
08000H	+	-----	+	.	(128
	.		.	.	bytes
	.		.	.	+
	.		.	.	-----
	.		.	.	07A80H
	.		.	.	+
	.		.	.	-----
	.		.	.	08000H
	.		.	.	
1FFFFH	+	-----	+		

A maximum of 4 pages may be used in the same manner.

A maximum of 4 pages may be allocated in the same manner (using a 128K-byte VRAM).

GRAPHIC 5 MODE

Characteristics

- Bit-mapped Graphics Mode
- Screen size : 512 (w) x 212 (h) dots
: 512 (w) x 192 (h) dots
- Screen colors : 4 colors out of 512 colors (per screen)
- Sprite mode : Sprite mode 2
- VRAM area per screen : 32K bytes

Controls

- Graphics : VRAM pattern name table
- Background color code : Low-order four bits of R#7
- Sprites : VRAM sprite attribute table, VRAM sprite pattern table

Initial Settings

1. Mode and Register Settings

	MSB	7	6	5	4	3	2	1	0	LSB								
	+---+---+---+---+---+---+---+---+---+---+																	
R#0		0		DG		IE2		IE1		1*		0*		0*		0		Mode register 0
	+===+===+===+===+===+===+===+===+===+===+																	
R#1		0		BL		IE0		0*		0*		0		SI		MAG		Mode register 1
	+===+===+===+===+===+===+===+===+===+===+																	
R#8		MS		LP		TP		CB		VR		0		SPD		BW		Mode register 2
	+===+===+===+===+===+===+===+===+===+===+																	
R#9		LN		0		S1		S0		IL		EO		**NT		DC		Mode register 3
	+---+---+---+---+---+---+---+---+---+---+																	

* Examples of settings in GRAPHIC 4 mode

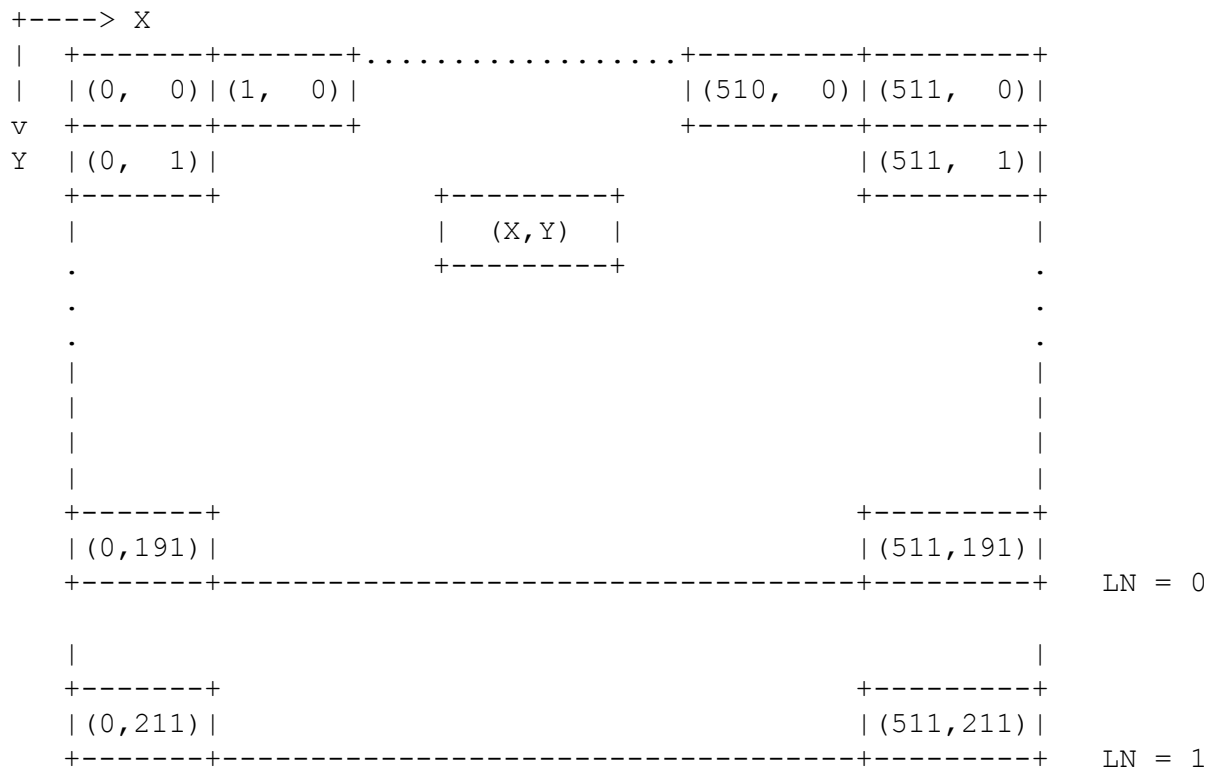
** Indicates negative logic

In GRAPHIC 5 mode, if LN is set to 1, the screen height is 212 dots, and if LN is set to 0, the screen height is 192 dots.

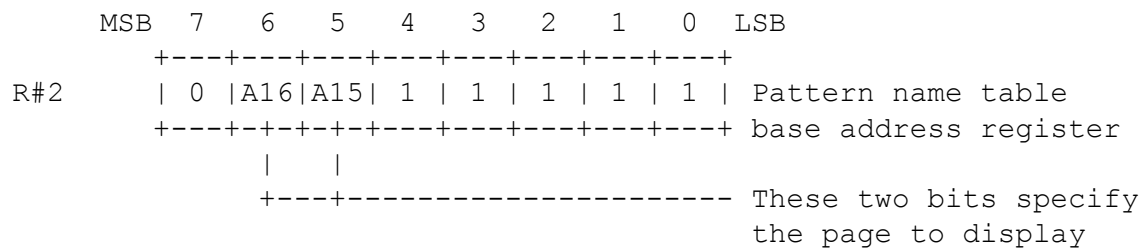
All other bits are set accordingly.

2. Pattern name table settings

- The pattern name table is composed of one byte for every four dots on the screen. A color can be assigned for each dot from a selection of 4 colors out of 512 colors.



- Set the beginning (head) address of the pattern name table in register R#2.



Pattern name table

MSB	7	6	5	4	3	2	1	0	LSB
	+---+---+---+---+---+---+---+---+								
1	(0, 0) (1, 0) (2, 0) (3, 0)								Base address
	+---+---+---+---+---+---+---+---+								
2	(4, 0) (5, 0) (6, 0) (7, 0)								Set the color code for each dot in this table
	+---+---+---+---+---+---+---+---+								
	.								.
	.								.
	.								.
	+---+---+---+---+---+---+---+---+								
127	(508,0) (509,0) (510,0) (511,0)								
	+---+---+---+---+---+---+---+---+								
128	(0, 1) (1, 1) (2, 1) (3, 1)								
	+---+---+---+---+---+---+---+---+								
	.								.
	.								.
	.								.
	+---+---+---+---+---+---+---+---+								
27135	(511,211)								
	+---+---+---+---+---+---+---+---+								

3. Color register settings

	MSB	7	6	5	4	3	2	1	0	LSB		
		+---+---+---+---+---+---+---+---+										
R#7		TC3	TC2	TC1	TC0	BD3	BD2	BD1	BD0		Text color/Back drop	
		+--+--+--+--+--+--+--+--+--+--+--+--+									color register	
						+---+---+---+---					Specifies backdrop color	
												code
		+---+---+---+-----									Ignored	

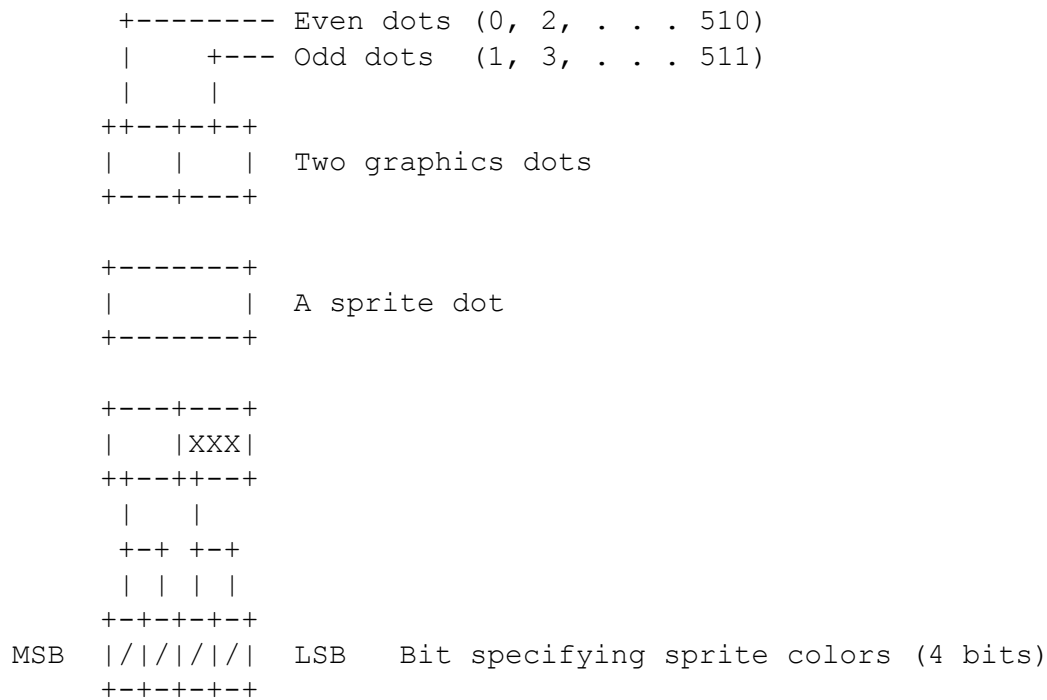
4. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 2.

	MSB	7	6	5	4	3	2	1	0	LSB	
R#5	+---+---+---+---+---+---+---+---+---+										
	A14	A13	A12	A11	A10	1	1	1			Sprite attribute table
	+===+===+===+===+===+===+===+===+===+										
R#11	0	0	0	0	0	0	A16	A15			base address register
	+---+---+---+---+---+---+---+---+---+										
	+---+---+---+---+---+---+---+---+---+										
R#6	0	0	A16	A15	A14	A13	A12	A11			Sprite pattern generator
	+---+---+---+---+---+---+---+---+---+										
											table base address
											register

5. Hardware tiling function

- In GRAPHIC 5 mode, a hardware tiling function processes the sprite and background colors. For these colors, you can specify four bits; however, of these four bits, the higher-order two bits specify the color code of the even dots, and the lower-order two bits specify the color code of the odd dots of the x-coordinage (0 to 511).
- In GRAPHIC 5 mode, the size of one dot of a sprite is approximately twice that of a graphics dot; however, when this tiling function is used, one dot of a sprite may be displayed in two colors.
- The even and odd dots of the background colors may also be specified in the same manner.



Example of VRAM allocation in GRAPHIC 5 mode

00000H	+-----+		
		Pattern	
		name	
02000H		table	
04000H			
		192 lines	
06000H	+-----+		
			.+-----+ 07000H
			Sprite
		212 lines	.- generator
06A00H	+-----+		table
			(2048
07000H	+-----+		bytes)
			+-----+ 07800H
			Sprite
			color
			table
			(512
			bytes)
			+-----+ 07A00H
			Sprite
			attribute
			table
			(128
			bytes
			+-----+ 07A80H
08000H	+-----+		+-----+ 08000H
.		.	
.		.	
.		.	A maximum of 4 p
.		.	the same manner
.		.	
.		.	
1FFFFH	+-----+		

A maximum of 4 pages may be allocated in the same manner (using a 128K-byte VRAM).

GRAPHIC 6 MODE

Characteristics

- Bit-mapped Graphics Mode
- Screen size : 512 (w) x 212 (h) dots
: 512 (w) x 192 (h) dots
- Screen colors : 16 colors out of 512 colors (per screen)
- Sprite mode : Sprite mode 2
- VRAM area per screen : 128K bytes (Two screens)

* To use this mode, the VRAM must have 128K bytes.

Controls

- Graphics : VRAM pattern name table
- Background color code : Low-order four bits of R#7
- Sprites : VRAM sprite attribute table, VRAM sprite pattern table

Initial Settings

1. Mode and Register Settings

	MSB	7	6	5	4	3	2	1	0	LSB	
R#0		0	DG	IE2 IE1	1*	0*	1*	0			Mode register 0
R#1		0	BL	IE0	0*	0*	0	SI	MAG		Mode register 1
R#8		MS	LP	TP	CB	VR	0	SPD	BW		Mode register 2
R#9		LN	0	S1	S0	IL	EO	**NT	DC		Mode register 3

* Examples of settings in GRAPHIC 4 mode

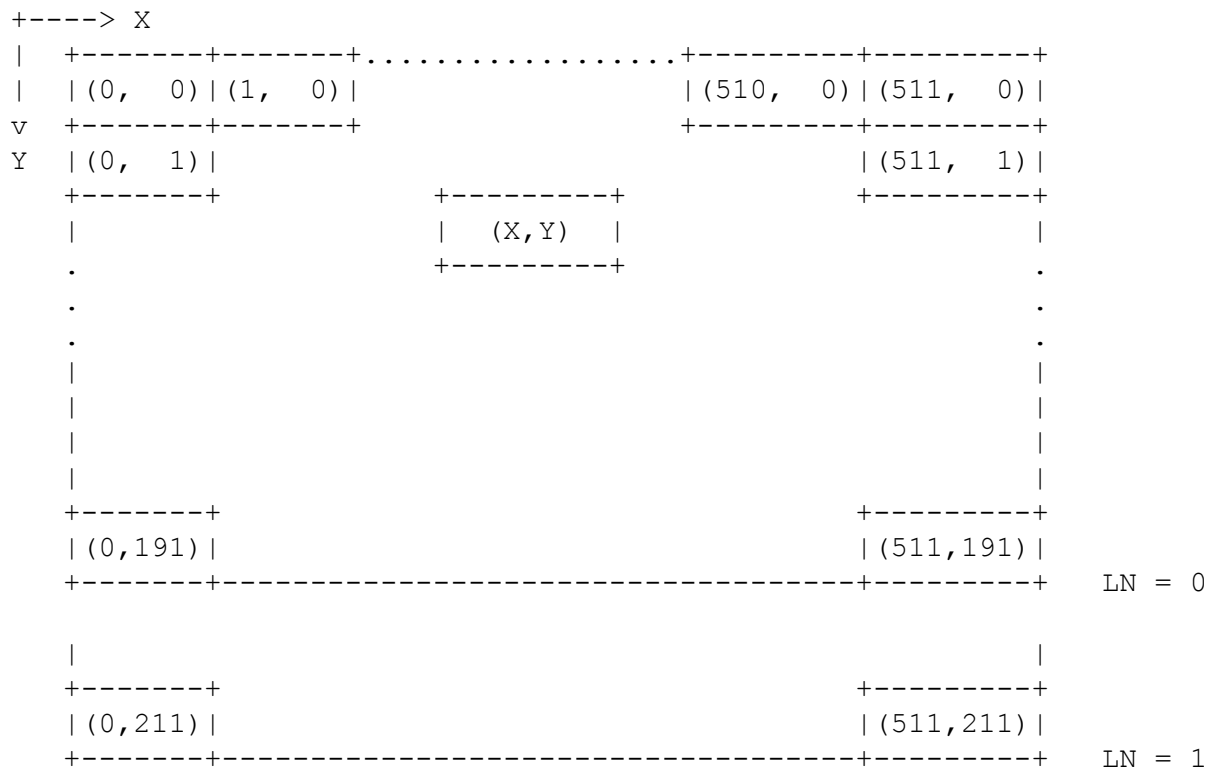
** Indicates negative logic

In GRAPHIC 6 mode, if LN is set to 1, the screen height is 212 dots, and if LN is set to 0, the screen height is 192 dots.

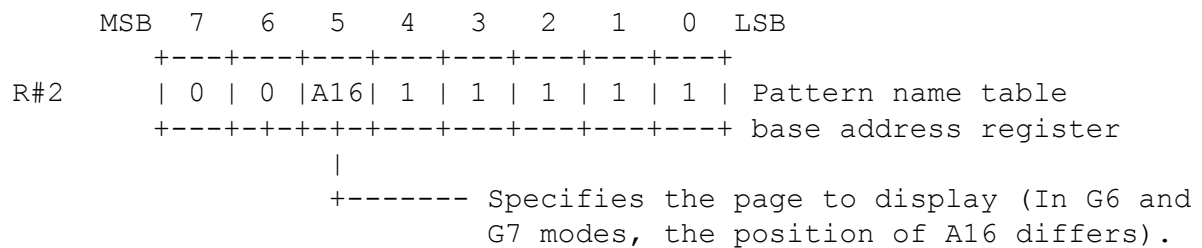
All other bits are set accordingly.

2. Pattern name table settings

- The pattern name table is composed of one byte for every two dots on the screen. A color can be assigned for each dot from a selection of 16 colors out of 512 colors.



- Set the beginning (head) address of the pattern name table in register R#2.



Pattern name table

MSB	7	6	5	4	3	2	1	0	LSB
	+---+---+---+---+---+---+---+---+								
1		(0, 0)				(1, 0)			Base address
	+---+---+---+---+---+---+---+---+								
2		(2, 0)				(3, 0)			Set the color code for each
	+---+---+---+---+---+---+---+---+								dot in this table
	.								.
	.								.
	.								.
	+---+---+---+---+---+---+---+---+								
255		(510, 0)				(511, 0)			
	+---+---+---+---+---+---+---+---+								
256		(0, 1)				(1, 1)			
	+---+---+---+---+---+---+---+---+								
	.								.
	.								.
	.								.
	+---+---+---+---+---+---+---+---+								
54270		(508,211)				(509,211)			
	+---+---+---+---+---+---+---+---+								
54271		(510,211)				(511,211)			
	+---+---+---+---+---+---+---+---+								

3. Color register settings

	MSB	7	6	5	4	3	2	1	0	LSB		
	+---+---+---+---+---+---+---+---+---+											
R#7		TC3	TC2	TC1	TC0	BD3	BD2	BD1	BD0		Text color/Back drop	
	+---+---+---+---+---+---+---+---+---+										color register	
						+---+---+---+---+---					Specifies backdrop color	
												code
	+---+---+---+---+---+---+---+---+---										Ignored	

4. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 2.

	MSB	7	6	5	4	3	2	1	0	LSB									
		+	---	+	---	+	---	+	---	+									
R#5			A14		A13		A12		A11		A10		1		1		1		Sprite attribute table
		+	===	+	===	+	===	+	===	+	===	+	===	+	===	+	===	+	
R#11			0		0		0		0		0		0		A16		A15		base address register
		+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	
		+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	
R#6			0		0		A16		A15		A14		A13		A12		A11		Sprite pattern generator
		+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	table base address
		+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	---	+	register

Example of VRAM allocation in GRAPHIC 6 mode

00000H	+-----+				
		Pattern			
		Name			
		table			
04000H					
08000H					
				+-----+	0F000H
				. Sprite	
		192 line		. generator	
0C000H	+-----+			. table	
		212 line		. (2048	
0D400H	+-----+			. bytes)	
				. +-----+	0F800H
				. Sprite	
				. color	
				. table	
				. (512	
				. bytes)	
				. +-----+	0FA00H
				. Sprite	
				. attribute	
				. table	
				. (128	
				. bytes	
0F000H	+-----+			+-----+	0FA80H
08000H	+-----+			+-----+	10000H
	.	.			
	.	.			
	.	.			
	.	.			
	.	.			
1FFFFH	+-----+				

A maximum of 2 pages may be allocated in the same manner (using a 128K-byte VRAM).

GRAPHIC 7 MODE

Characteristics

- Bit-mapped Graphics Mode
- Screen size : 256 (w) x 212 (h) dots
: 256 (w) x 192 (h) dots
- Screen colors : 256 colors (per screen)
- Sprite mode : Sprite mode 2
- VRAM area per screen : 128K bytes (Two screens)

* To use this mode, the VRAM must have 128K bytes.

Controls

- Graphics : VRAM pattern name table
- Background color code : Low-order four bits of R#7
- Sprites : VRAM sprite attribute table, VRAM sprite pattern table

Initial Settings

1. Mode and Register Settings

	MSB	7	6	5	4	3	2	1	0	LSB	
R#0		0	DG	IE2 IE1	1*	1*	1*	0			Mode register 0
R#1		0	BL	IE0	0*	0*	0	SI	MAG		Mode register 1
R#8		MS	LP	TP	CB	VR	0	SPD BW			Mode register 2
R#9		LN	0	S1	S0	IL	EO	**NT DC			Mode register 3

* Examples of settings in GRAPHIC 4 mode

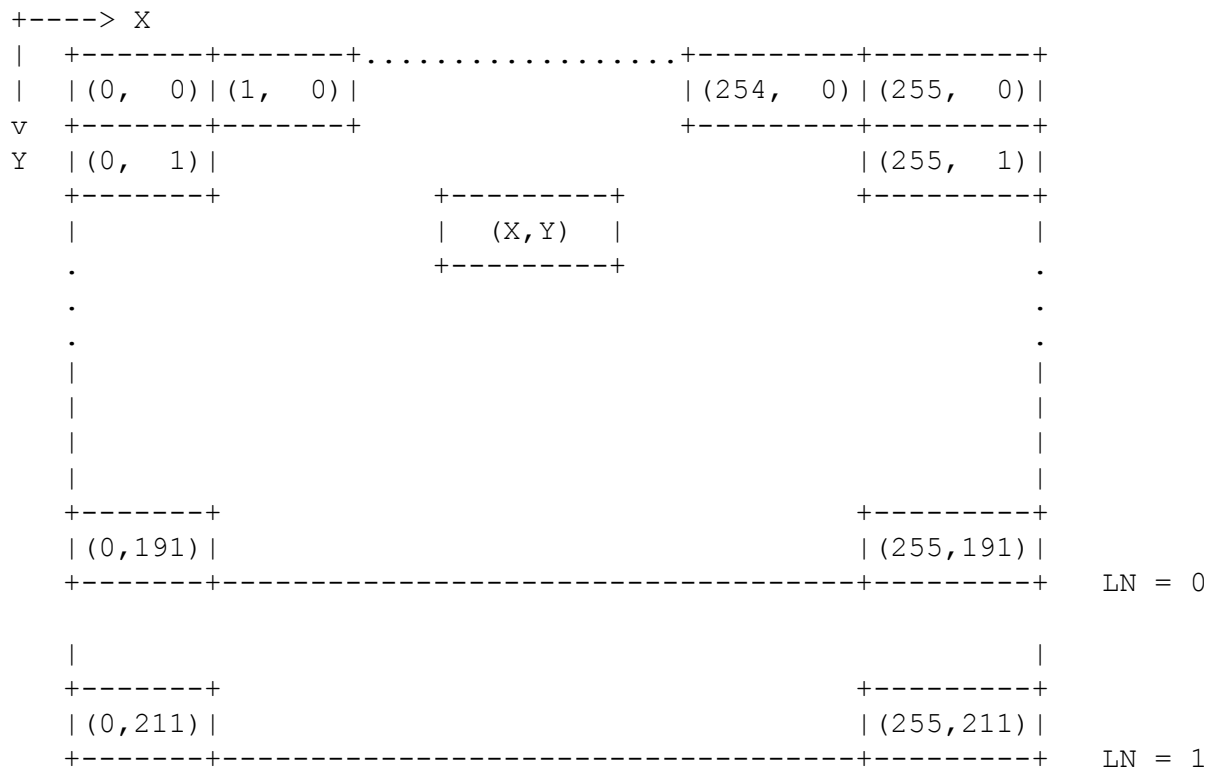
** Indicates negative logic

In GRAPHIC 7 mode, if LN is set to 1, the screen height is 212 dots, and if LN is set to 0, the screen height is 192 dots.

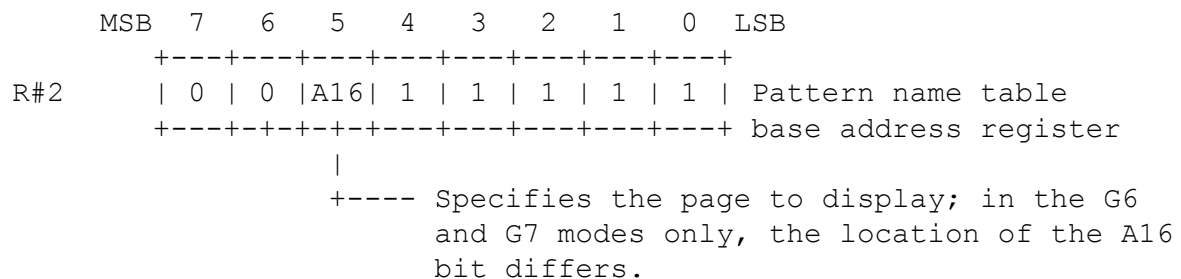
All other bits are set accordingly.

2. Pattern name table settings

- The pattern name table is composed of one byte for every dot on the screen. A color can be assigned for each dot from a selection of 256 colors.



- Set the beginning (head) address of the pattern name table in register R#2.



Pattern name table

MSB	7	6	5	4	3	2	1	0	LSB
1				(0, 0)					Base address
2				(1, 0)					Set the color code for each dot in this table
.									.
		GREEN		RED		BLUE			
255				(255, 0)					
256				(0, 1)					
.									.
.									.
.									.
54270				(254, 211)					
54271				(255, 211)					

3. Color register settings

MSB	7	6	5	4	3	2	1	0	LSB
R#7									
		TC3		TC2		TC1		TC0	
		BD3		BD2		BD1		BD0	

Text color/Back drop color register

Specifies backdrop color code

4. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 2.

	MSB	7	6	5	4	3	2	1	0	LSB		
R#5		+	-	-	+	-	-	+	-	-	+	Sprite attribute table
			A14		A13		A12		A11		A10	
		+	=	=	+	=	=	+	=	=	+	
R#11			0		0		0		0		0	base address register
		+	-	-	+	-	-	+	-	-	+	
		+	-	-	+	-	-	+	-	-	+	
R#6			0		0		A16		A15		A14	Sprite pattern generator
		+	-	-	+	-	-	+	-	-	+	table base address
		+	-	-	+	-	-	+	-	-	+	register

Example of VRAM allocation in GRAPHIC 7 mode

00000H	+-----+				
		Pattern			
		Name			
		table			
04000H					
08000H					
				+-----+	0F000H
				. Sprite	
		192 line		. generator	
0C000H	+-----+			. table	
		212 line		. (2048	
0D400H	+-----+			. bytes)	
				. +-----+	0F800H
				. Sprite	
				. color	
				. table	
				. (512	
				. bytes)	
				. +-----+	0FA00H
				. Sprite	
				. attribute	
				. table	
				. (128	
				. bytes	
0F000H	+-----+			+-----+	0FA80H
08000H	+-----+			+-----+	10000H
		.			
		.			
		.			
		.			
		.			
		.			
1FFFFH	+-----+				

A maximum of 2 pages may be allocated in the same manner (using a 128K-byte VRAM).

COMMANDS

1. Types of Commands

It is very easy to use MSX-VIDEO commands to perform functions such as LINE and PSET for use with graphics, and for transferring parts of the screen.

Command Name	Destination	Source	Unit	Mnemonic	CM3	CM2	CM1	CM0
High-speed move	VRAM	CPU	Byte	HMMC	1	1	1	1
	VRAM	VRAM	Byte	YMMM	1	1	1	0
	VRAM	VRAM	Byte	HMMM	1	1	0	1
	VRAM	VDP	Byte	HMMV	1	1	0	0
Logical move	VRAM	CPU	Dot	LMMC	1	0	1	1
	CPU	VRAM	Dot	LMCM	1	0	1	0
	VRAM	VRAM	Dot	LMMM	1	0	0	1
	VRAM	VDP	Dot	LMMV	1	0	0	0
Line	VRAM	VDP	Dot	LINE	0	1	1	1
Search	VRAM	VDP	Dot	SRCH	0	1	1	0
Pset	VRAM	VDP	Dot	PSET	0	1	0	1
Point	VDP	VRAM	Dot	POINT	0	1	0	0
Invalid					0	0	1	1
Invalid					0	0	1	0
Invalid					0	0	0	1
Stop					0	0	0	0

- Commands are executed in the MSX-VIDEO by writing the data into R#46 (the Command Register, hereafter abbreviated CMR), and setting bit 0 of status register S#2 (CE/Command Execute) to 1. Before this can be done, the necessary parameters must first have been set in registers R#32 to R#45.
- When the command execution is complete, CE is set to 0.
- To abort a command while it is being executed, execute a STOP.
- The results of command execution are only guaranteed during bit map mode (Graph4 to Graph7).

2. Page Concept

The parameters used for the MSX-VIDEO are all x-y coordinates. In other words, the internal command processor of the MSX-VIDEO accesses the entire VRAM area as x-y coordinates of the display mode.

When a screen is to be displayed, 212 lines of the same page are displayed (selected by R#23). To select the page to be displayed, use R#2.

When a command is being executed, the contents of the display screen are ignored.

The display modes and their relationships to the coordinates are shown in the table below.

GRAPH 4	Address	GRAPH 5
+-----+ (0,0) (255,0) Page 0 (0,255) (255,255) +-----+	00000H +-----+	+-----+ (0,0) (511,0) Page 0 (0,255) (511,255) +-----+
+-----+ (0,256) (255,256) Page 1 (0,511) (255,511) +-----+	08000H +-----+	+-----+ (0,256) (511,256) Page 1 (0,511) (511,511) +-----+
+-----+ (0,512) (255,512) Page 2 (0,767) (255,767) +-----+	10000H +-----+	+-----+ (0,512) (511,512) Page 2 (0,767) (511,767) +-----+
+-----+ (0,768) (255,768) Page 3 (0,1023) (255,1023) +-----+	18000H +-----+	+-----+ (0,768) (511,768) Page 3 (0,1023) (511,1023) +-----+
	1FFFFH	
GRAPH 7	Address	GRAPH 6
+-----+ (0,0) (255,0) Page 0 (0,255) (255,255) +-----+	00000H +-----+	+-----+ (0,0) (511,0) Page 0 (0,255) (511,255) +-----+
+-----+ (0,256) (255,256) Page 1 (0,511) (255,511) +-----+	10000H +-----+	+-----+ (0,256) (511,256) Page 1 (0,511) (511,511) +-----+
	1FFFFH	

3. Logical Operations

When the LINE, PSET and LOGICAL MOVE commands are executed on the MSX-VIDEO, the operations may be performed on the color on the screen. To do logical operations on the MSX-VIDEO, write the lower four bits of R#46 (Command register) simultaneously when you specify the command.

Summary of Logical Operations

Name	Operation	LO3	LO2	LO1	LO0
IMP	DC=SC	0	0	0	0
AND	SC*DC	0	0	0	1
OR	SC+DC	0	0	1	0
EOR	$\overline{SC*DC} + SC*\overline{DC}$	0	0	1	1
NOT	DC= $\overline{SC}$	0	1	0	0
---		0	1	0	1
---		0	1	1	0
---		0	1	1	1
TIMP	if SC=0 then DC=DC else DC=SC	1	0	0	0
TAND	if SC=0 then DC=DC else SC*DC	1	0	0	1
TOR	if SC=0 then DC=DC else SC+DC	1	0	1	0
TEOR	if SC=0 then DC=DC else $\overline{SC*DC} + SC*\overline{DC}$	1	0	1	1
TNOT	if SC=0 then DC=DC else DC= $\overline{SC}$	1	1	0	0
---		1	1	0	1
---		1	1	1	0
---		1	1	1	1

* SC = Source Color code

* DC = Destination Color code

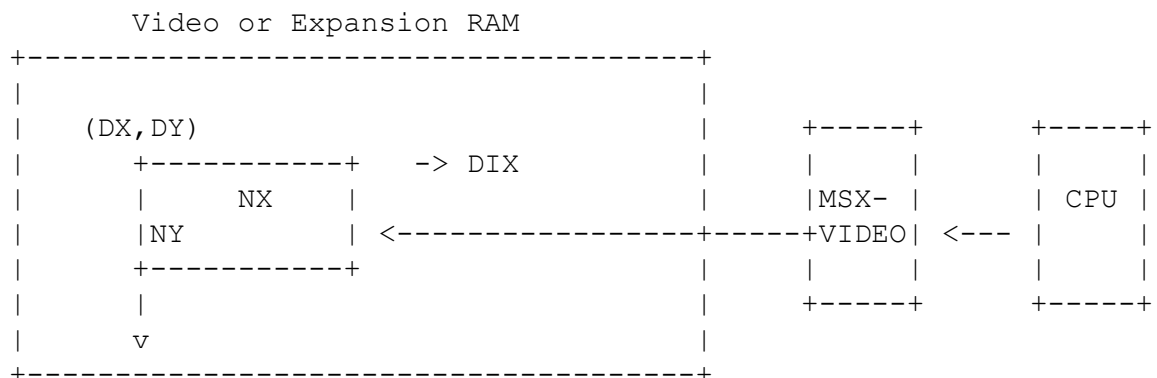
* EOR = Exclusive OR

4. Explanation of Commands

4.1 HMMC (High-speed move CPU to VRAM)

The HMMC command transfers data from the CPU to the Video or expansion RAM in a specified rectangular area (in x-y coordinates) via the MSX-VIDEO.

Since the data to be transferred is done in units of one byte, there is a limitation, according to the display mode, on the value for x.



4.1.1 HMMC Execution Order

1. First, set the necessary parameters in the command register of the MSX-VIDEO.

MXD: Select destination memory

0: Video RAM

1: Expansion RAM

DX: Basic x-coordinate of destination (0 to 511) *1

DY: Basic y-coordinate of destination (0 to 1023)

NX: Dots to move in x-direction (0 to 511) *1

NY: Dots to move in y-direction (0 to 1023)

*1 Note that in the G4 and G6 modes, the lower one bit, and in the G5 mode, the lower two bits are lost.

DIX: Direction for NX from x-coordinate of destination

0: Right

1: Left

DIY: Direction for NY from y-coordinate of destination

0: Down

1: Up

CLR: (R#44: Color register)

First byte of data to be transferred

2. After you specify the above data, execute the command by writing 1 1 1 1 0 0 0 0 B into the CMR (R#46: Command register).
3. While checking TR and CE in Status Register #2, send the second byte and all bytes following into the CLR register.

4.1.2 Setting up the HMMC register